

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```

nan_ratio = df[cols_to_check].isna().mean()
valid_features = nan_ratio[nan_ratio <= 0.15].index
valid_features_list = list(valid_features)

#df = df[valid_features]
# Removing of last np.nan
mask = np.isfinite(df[valid_features]).all(axis=1)
df = df[mask]

cols = list(dict.fromkeys(valid_features_list + #target_col + features_X +
                        feats_to_balance + feats_grouping +
                        feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

new_feats_X = [x for x in features_X if x in valid_features_list]
new_feats_add = [x for x in feats_eng if x in valid_features_list]

data["df"]=df
data["features_X"] = new_feats_X
data["feats_added"] = new_feats_add

column_removed = [x for x in cols_to_check if x not in valid_features_list]
print(f'Column removed because np.isnan > 15%: {column_removed}')

return data

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":

```

```

df["campaign"] = df["datetime"].apply(assign_campaign)
df = df[df["campaign"] != "other"].copy()
df["innova_point"] = df.apply(assign_innova_point, axis=1)
df = df[df["innova_point"] != "no"].copy()
with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r") as f:
    #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["columnn"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}

```

```

        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")

```

Column removed because np.isnan > 15%: []

1.2 Metadata

Nombre de lignes df	7698
Nombre de lignes df clean	7698
Nombre de features	32
% de NaN	0.0
Taille dataset entraînement	5388
Taille dataset test	2309
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_D1_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_NH3_wl_12h_A_energy_rel, r_temperature_wl_12h_D1_energy, r_temperature_wl_12h_D2_energy, r_temperature_wl_12h_D3_energy, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_temperature_wl_12h_D1_energy_rel, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy_rel, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_D2_energy, r_umidity_wl_12h_D3_energy, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D3_energy_rel, r_umidity_wl_12h_A_energy_rel

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

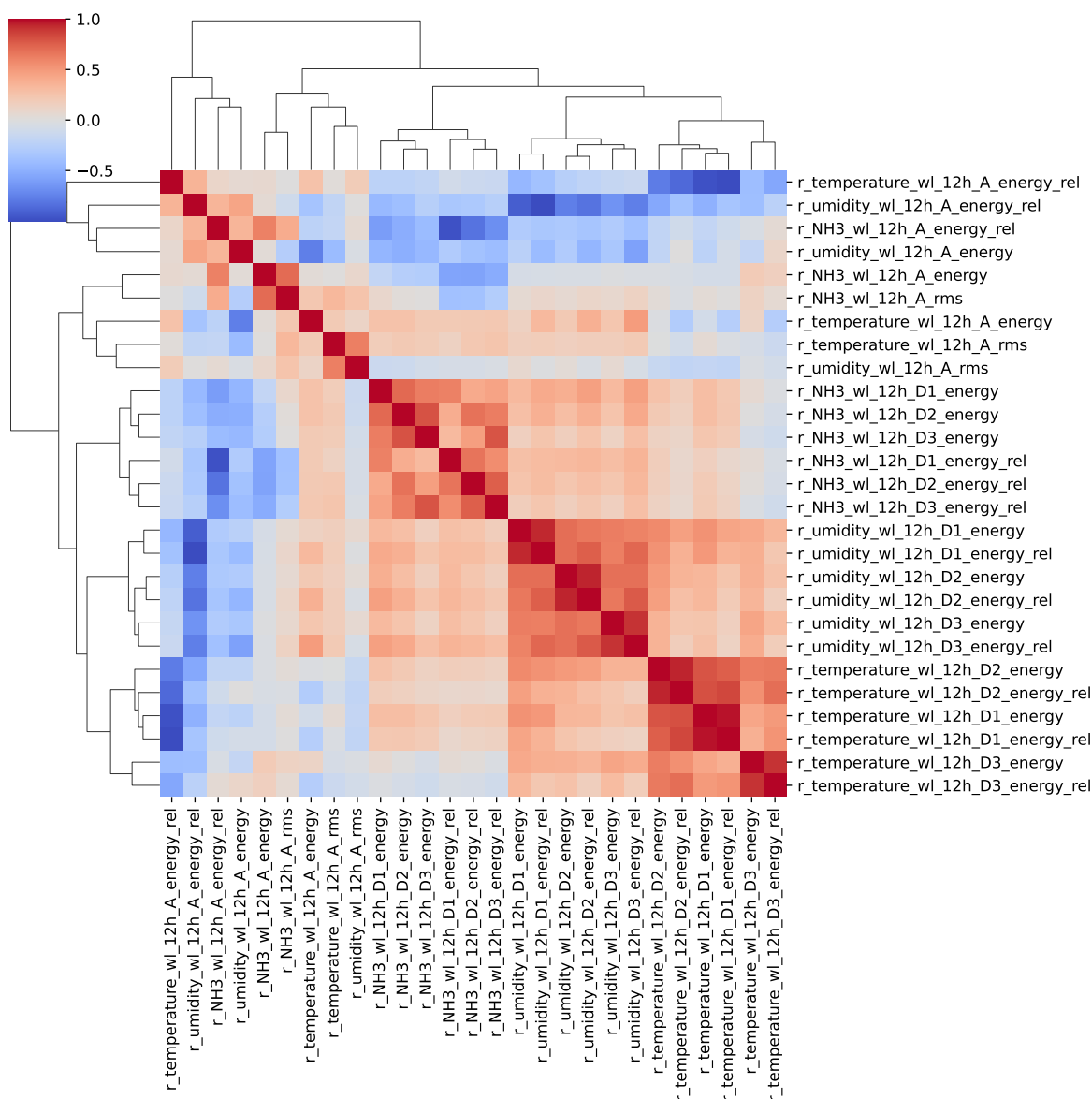
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, , N, H, 3 ; **cluster 2:** r, , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, , u, m, i, d, i, t, y ; **cluster 4:** r, , C, O, 2 ; **cluster 5:** r, , T, H, I ; **cluster 6:** r_NH3_wl_12h_D1_energy ; **cluster 7:** r_NH3_wl_12h_D2_energy ; **cluster 8:** r_NH3_wl_12h_D3_energy ; **cluster 9:** r_NH3_wl_12h_A_energy ; **cluster 10:** r_NH3_wl_12h_A_rms ; **cluster 11:** r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D2_energy_rel, r_NH3_wl_12h_D1_energy_rel ; **cluster 12:** r_NH3_wl_12h_D3_energy_rel ; **cluster 13:** r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D1_energy_rel, r_temperature_wl_12h_D1_energy, r_temperature_wl_12h_D2_energy_rel, r_temperature_wl_12h_D2_energy ; **cluster 14:** r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_D3_energy ; **cluster 15:** r_temperature_wl_12h_A_energy ; **cluster 16:** r_temperature_wl_12h_A_rms ; **cluster 17:** r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D2_energy_rel, r_umidity_wl_12h_D1_energy, r_umidity_wl_12h_A_energy_rel, r_umidity_wl_12h_D2_energy ; **cluster 18:** r_umidity_wl_12h_D3_energy_rel, r_umidity_wl_12h_D3_energy ; **cluster 19:** r_umidity_wl_12h_A_energy ; **cluster 20:** r_umidity_wl_12h_A_rms ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D3_energy_rel, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_wl_12h_D1_energy, r_NH3_wl_12h_D2_energy, r_NH3_wl_12h_D3_energy, r_NH3_wl_12h_A_energy, r_NH3_wl_12h_A_rms, r_NH3_wl_12h_A_energy_rel, r_NH3_wl_12h_D3_energy_rel,

r_temperature_wl_12h_A_energy_rel, r_temperature_wl_12h_D3_energy_rel, r_temperature_wl_12h_A_energy, r_temperature_wl_12h_A_rms, r_umidity_wl_12h_D1_energy_rel, r_umidity_wl_12h_D3_energy_rel, r_umidity_wl_12h_A_energy, r_umidity_wl_12h_A_rms

Len(Features_X): 20

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

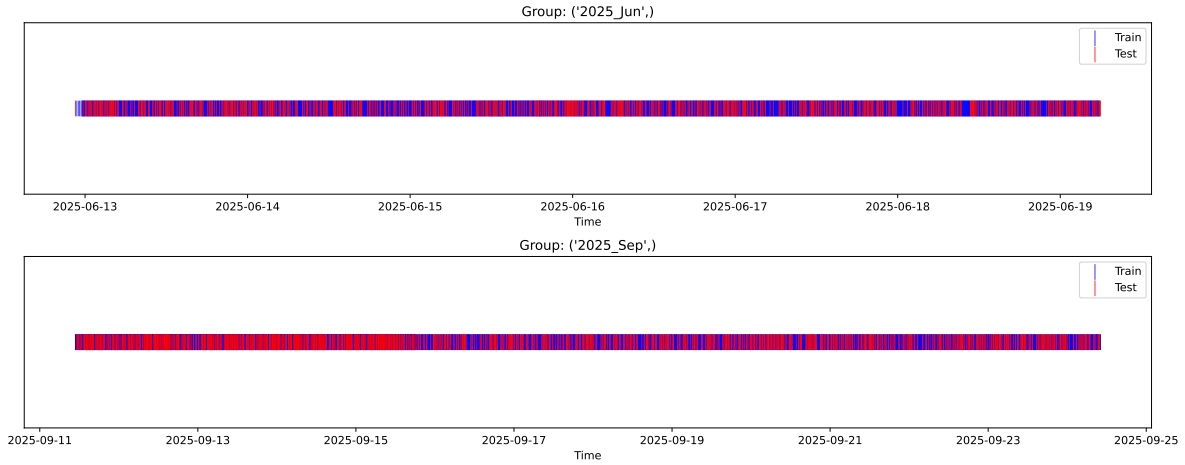
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

External Split: Train: 5388 samples Test: 2310 samples



3.3.2 Internal CV Folds

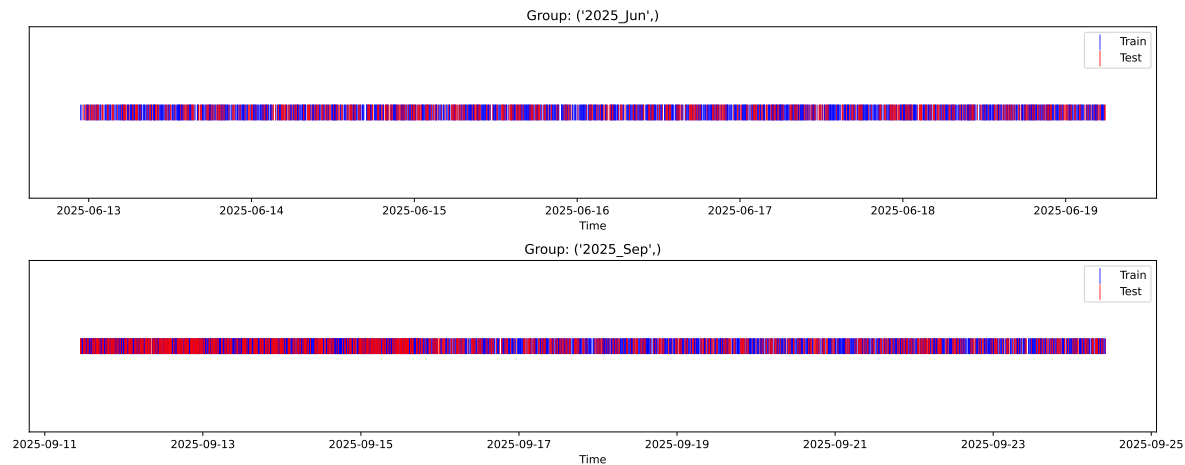
Fold 1/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



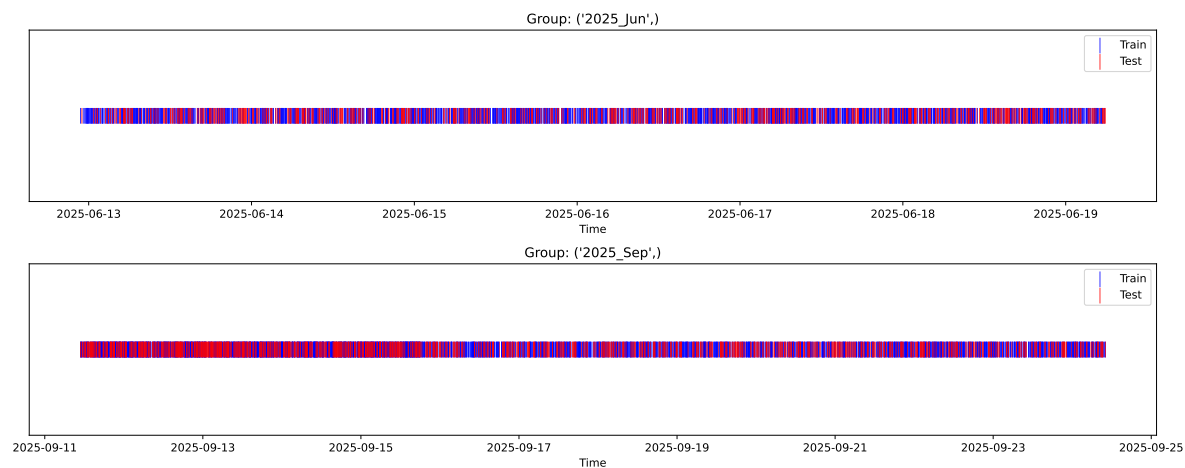
Fold 2/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



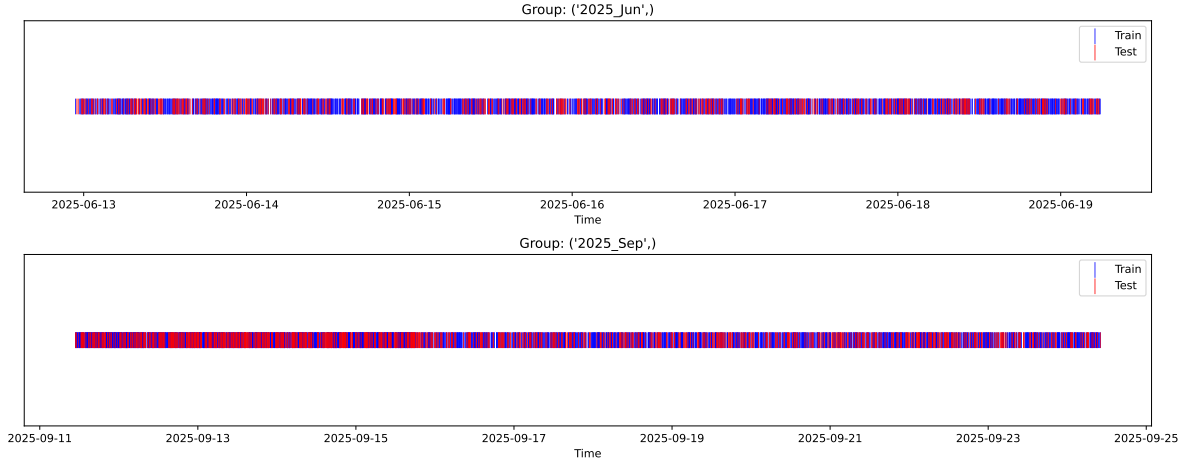
Fold 3/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



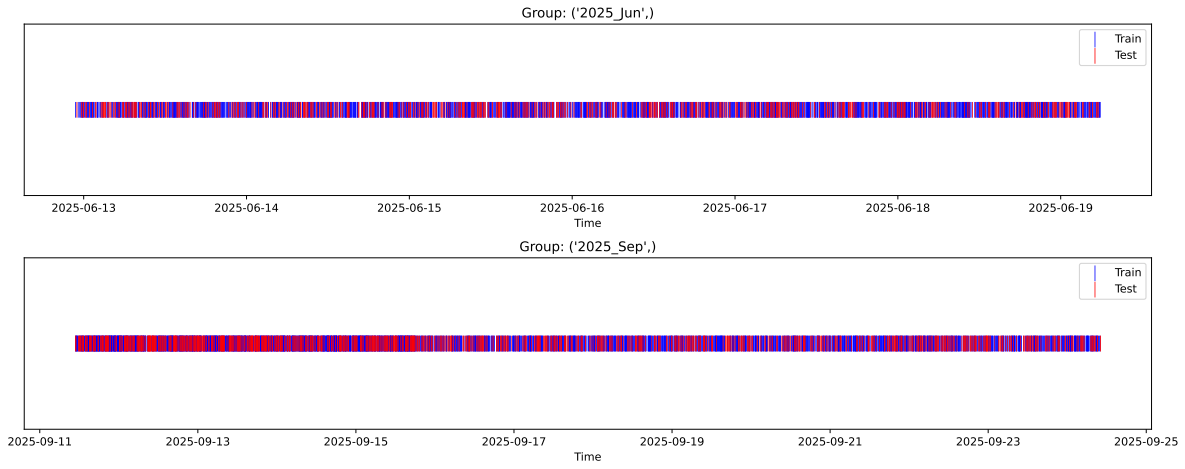
Fold 4/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3771 samples Test: 1617 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.347, std=0.812, min=0.959, max=13.845
 y_{test} : mean=2.347, std=0.772, min=1.011, max=7.779

Fold 0 — y_{test} : mean=2.39, std=0.85, y_{train} : mean=2.33, std=0.79

Fold 1 — y_{test} : mean=2.33, std=0.80, y_{train} : mean=2.36, std=0.82

Fold 2 — y_{test} : mean=2.32, std=0.80, y_{train} : mean=2.36, std=0.81

Fold 3 — y_test: mean=2.36, std=0.85, y_train: mean=2.34, std=0.80

Fold 4 — y_test: mean=2.35, std=0.80, y_train: mean=2.34, std=0.82

train scores CV: [0.51384726 0.51174903 0.51509823 0.51766316 0.5132031]

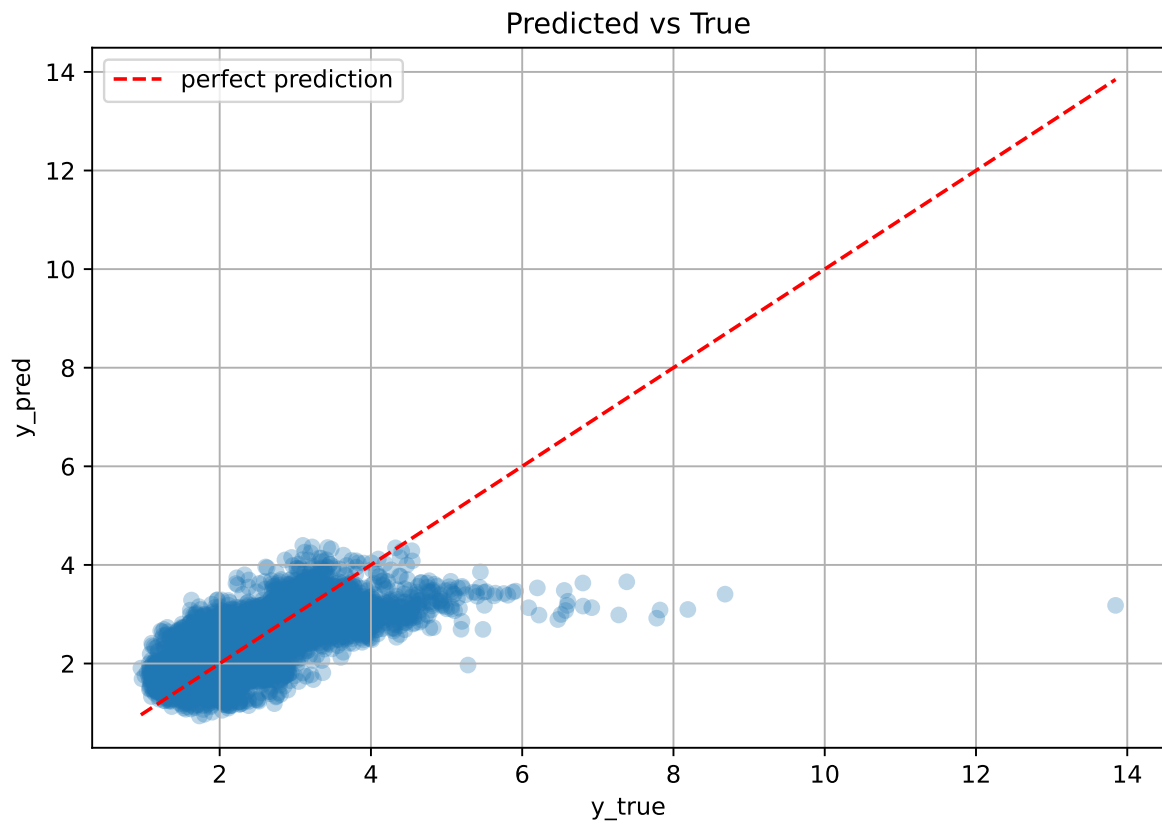
test scores CV: [0.50230087 0.51075259 0.5026405 0.4980221 0.50686063]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5143	0.5041	0.0043	0.5084	0.5086	0.0045	1.0089	1.0202

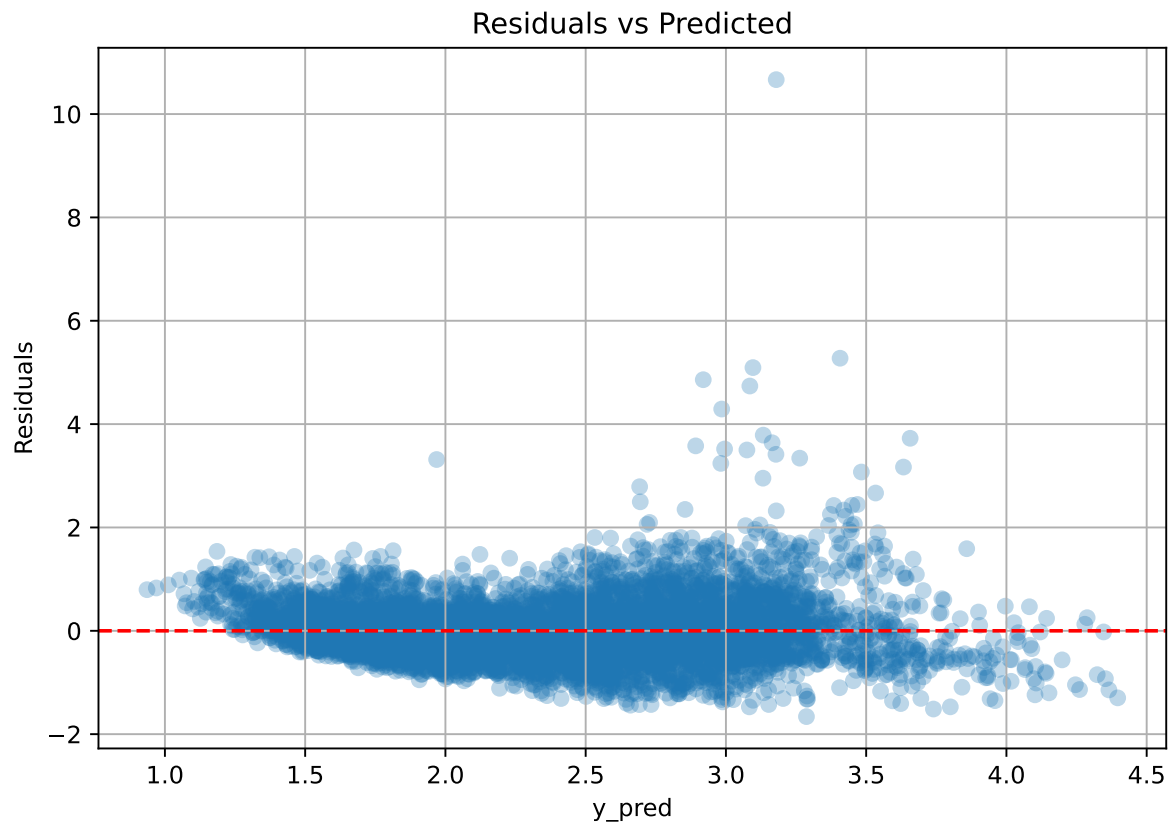
Coefficients:

coef_r_CO2 = -0.0628, coef_r_NH3 = 0.1223, coef_r_NH3_wl_12h_A_energy = -0.1630, coef_r_NH3_wl_12h_A_energy_rel = 0.0136, coef_r_NH3_wl_12h_A_rms = -0.0025, coef_r_NH3_wl_12h_D1_energy = 0.0550, coef_r_NH3_wl_12h_D2_energy = 0.0712, coef_r_NH3_wl_12h_D3_energy = -0.0311, coef_r_NH3_wl_12h_D3_energy_rel = -0.0461, coef_r_THI = 0.5758, coef_r_temperature = -0.7050, coef_r_temperature_wl_12h_A_energy = -0.0600, coef_r_temperature_wl_12h_A_energy_rel = -0.1443, coef_r_temperature_wl_12h_A_rms = -0.0837, coef_r_temperature_wl_12h_D3_energy_rel = -0.0858, coef_r_umidity = -0.6875, coef_r_umidity_wl_12h_A_energy = 0.1484, coef_r_umidity_wl_12h_A_rms = -0.1081, coef_r_umidity_wl_12h_D1_energy_rel = -0.0437, coef_r_umidity_wl_12h_D3_energy_rel = 0.0743, intercept = 2.3473,

3.4.2 Scatter



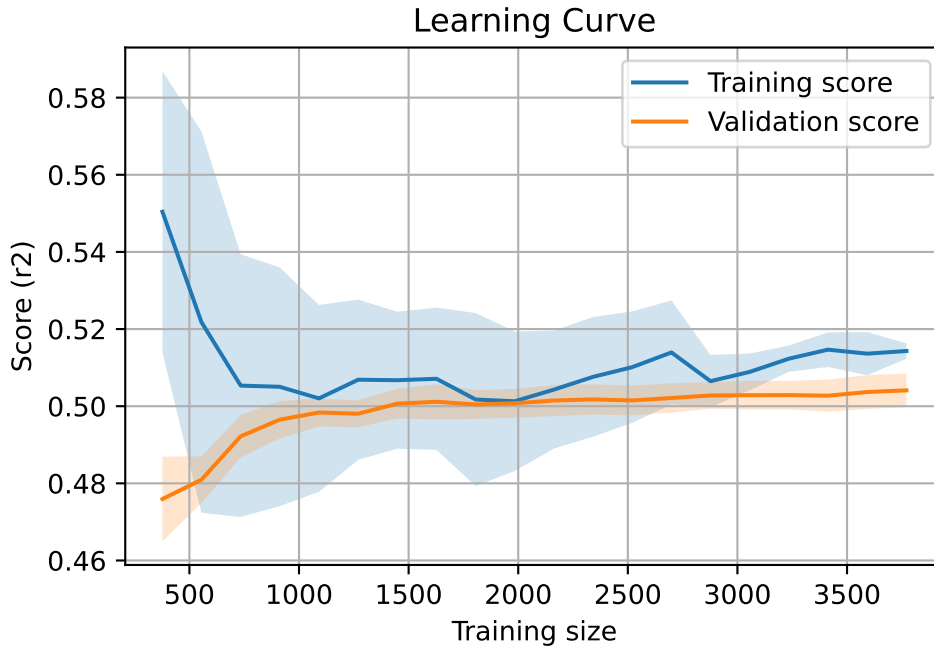
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7698, `Len(training)` : 5389,

`Len(training_real)` : 3773, `Len(test_of_training)` : 1616, `Len(test)` : 2309



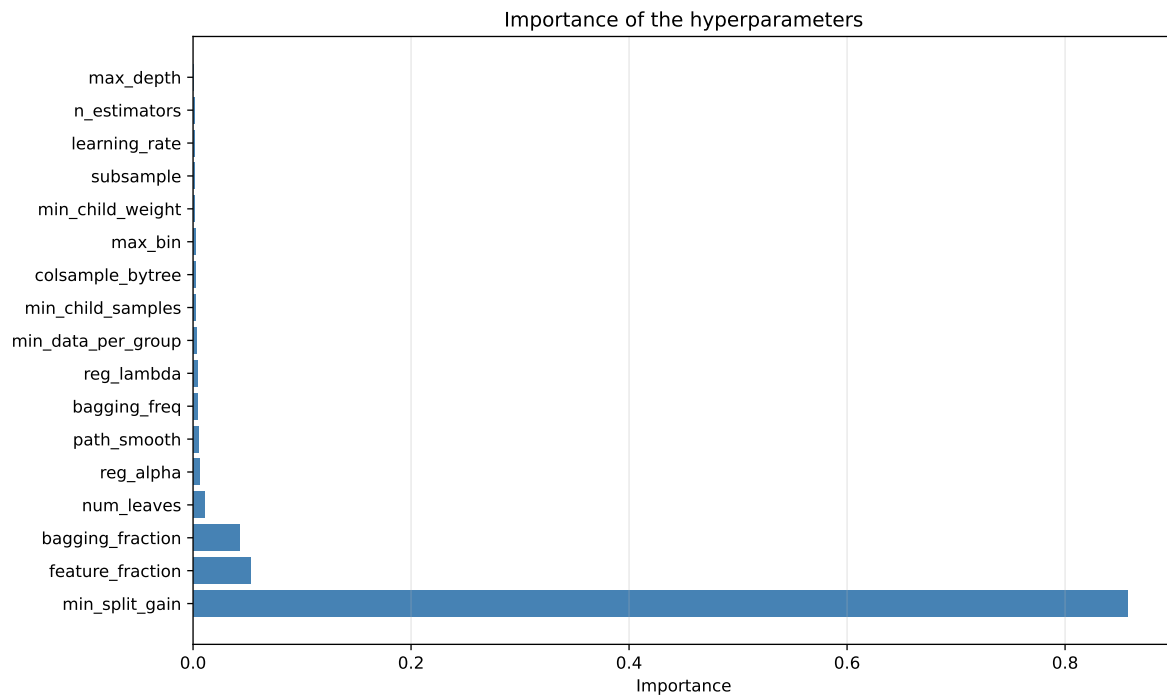
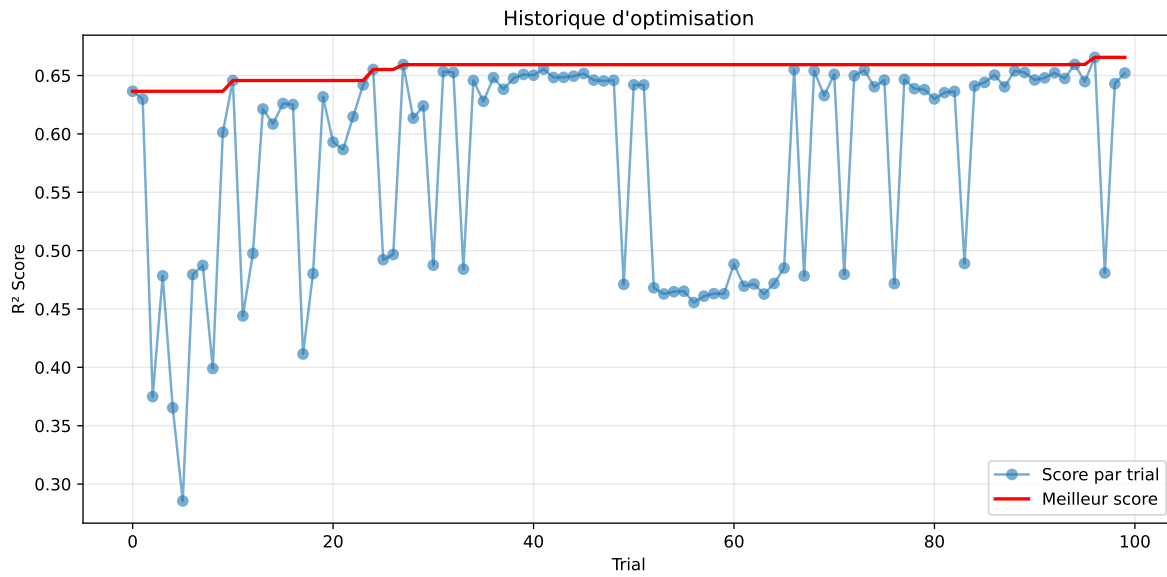
3.5 Model : LGBM

3.5.1 Gridsearch

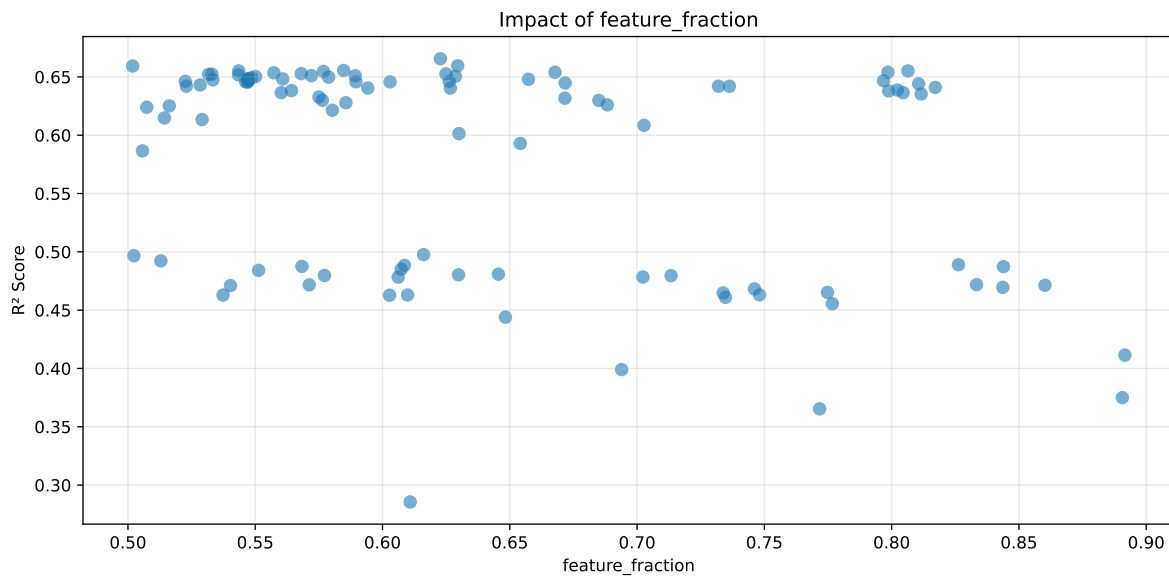
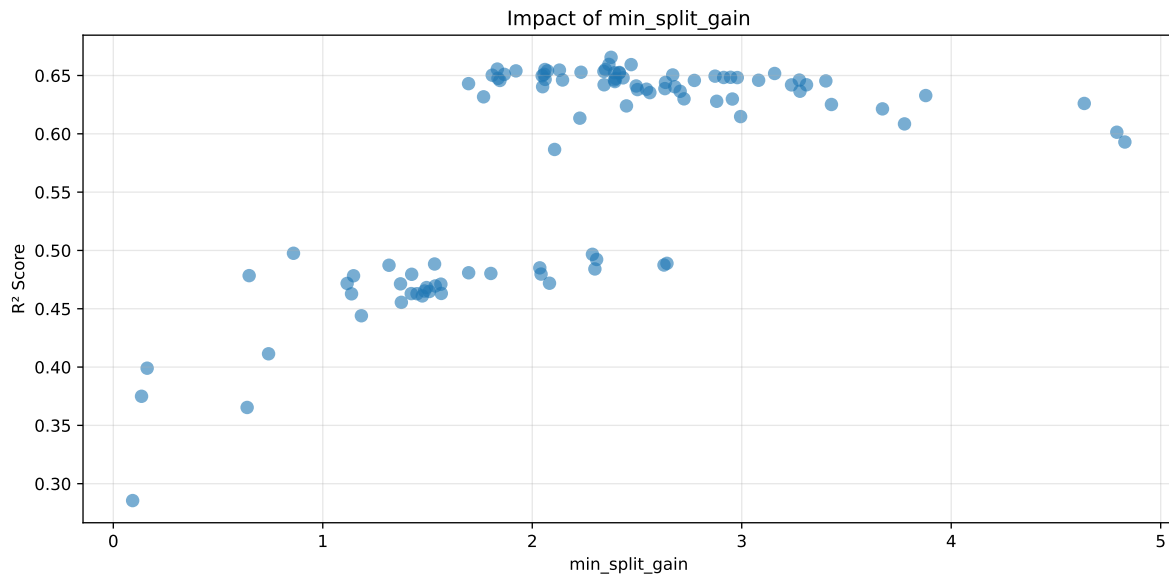
Distribution y_{train} vs y_{test} : y_{train} : mean=2.347, std=0.812, min=0.959, max=13.845
 y_{test} : mean=2.347, std=0.772, min=1.011, max=7.779

===== TOP Importance FEATURES =====

feature	importance
r_CO2	210
r_umidity	179
r_temperature_wl_12h_A_energy	174
r_temperature_wl_12h_A_rms	137
r_NH3	103
r_NH3_wl_12h_D1_energy	88
r_NH3_wl_12h_D3_energy_rel	85
r_umidity_wl_12h_A_rms	84
r_temperature_wl_12h_A_energy_rel	79
r_temperature	75
r_NH3_wl_12h_A_energy	71
r_umidity_wl_12h_A_energy	58
r_umidity_wl_12h_D3_energy_rel	51
r_NH3_wl_12h_D3_energy	44
r_THI	39
r_NH3_wl_12h_D2_energy	36
r_NH3_wl_12h_A_energy_rel	34
r_umidity_wl_12h_D1_energy_rel	32
r_NH3_wl_12h_A_rms	25
r_temperature_wl_12h_D3_energy_rel	11

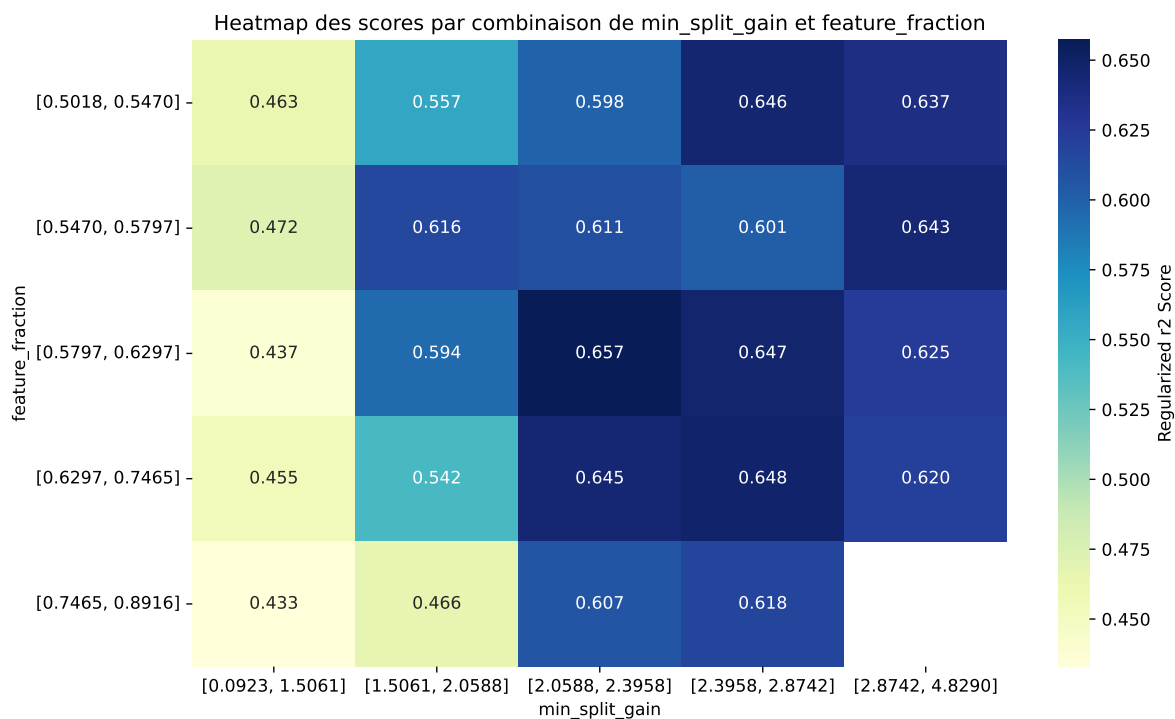


Paramètres avec >5% d'importance : ['min_split_gain', 'feature_fraction']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.697	0.6656	0.6707	0.0187	0.6656	0.6927	0.0271	1.0408	1.0471
2	0.6924	0.6595	0.6644	0.019	0.6595	0.6825	0.023	1.0349	1.0499
3	0.6909	0.6593	0.664	0.0191	0.6593	0.6858	0.0265	1.0402	1.0479
4	0.6882	0.6555	0.6608	0.0191	0.6555	0.685	0.0295	1.045	1.0499
5	0.6848	0.6552	0.6612	0.018	0.6552	0.682	0.0268	1.0409	1.0452
6	0.6857	0.6551	0.6604	0.0142	0.6551	0.6821	0.027	1.0412	1.0467
7	0.6849	0.6546	0.6593	0.0174	0.6546	0.679	0.0244	1.0373	1.0462
8	0.6862	0.654	0.6597	0.0212	0.654	0.6826	0.0286	1.0437	1.0492
9	0.6832	0.654	0.6588	0.0184	0.654	0.678	0.024	1.0367	1.0447
10	0.6857	0.6535	0.6593	0.0177	0.6535	0.6866	0.0331	1.0507	1.0493
---	---	---	---	---	---	---	---	---	---
100	0.831	0.7401	0.7548	0.0265	0.2855	0.7661	0.0261	1.0352	1.1228

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.029114101471965657, 'reg__num_leaves': 31, 'reg__subsample': 0.8343798331547473, 'reg__colsample_bytree': 0.8979736378732517, 'reg__min_child_samples': 39, 'reg__reg_alpha': 0.8889128412167562, 'reg__reg_lambda': 26.909286230377276, 'reg__min_split_gain': 2.3762082624595133, 'reg__path_smooth': 1.8445996293228974, 'reg__min_child_weight':

0.0012474315477679575, 'reg__feature_fraction': 0.6227447900554351, 'reg__bagging_fraction': 0.5210489617728931, 'reg__bagging_freq': 1, 'reg__max_bin': 74, 'reg__min_data_per_group': 180}

Rank 2: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.029346941991425544, 'reg__num_leaves': 30, 'reg__subsample': 0.7616538806893736, 'reg__colsample_bytree': 0.891472647517288, 'reg__min_child_samples': 18, 'reg__reg_alpha': 4.0326627148679055, 'reg__reg_lambda': 26.314166121473686, 'reg__min_split_gain': 2.365865308334709, 'reg__path_smooth': 3.0672138946715557, 'reg__min_child_weight': 0.0021582207864145717, 'reg__feature_fraction': 0.6295248447315074, 'reg__bagging_fraction': 0.6419786132347486, 'reg__bagging_freq': 1, 'reg__max_bin': 163, 'reg__min_data_per_group': 188}

Rank 3: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.013953704772623957, 'reg__num_leaves': 33, 'reg__subsample': 0.7614156089832588, 'reg__colsample_bytree': 0.641942569276627, 'reg__min_child_samples': 50, 'reg__reg_alpha': 0.0021503180465294314, 'reg__reg_lambda': 28.29916257046837, 'reg__min_split_gain': 2.4720304327940457, 'reg__path_smooth': 2.8914664987689664, 'reg__min_child_weight': 0.0014459347540684282, 'reg__feature_fraction': 0.5018448407809841, 'reg__bagging_fraction': 0.5013901747919409, 'reg__bagging_freq': 1, 'reg__max_bin': 175, 'reg__min_data_per_group': 195}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.019733915656416915, 'reg__num_leaves': 38, 'reg__subsample': 0.811695172889724, 'reg__colsample_bytree': 0.7021875643167605, 'reg__min_child_samples': 29, 'reg__reg_alpha': 5.363752627268926, 'reg__reg_lambda': 25.32764731719096, 'reg__min_split_gain': 1.833350875429082, 'reg__path_smooth': 1.8431729510069998, 'reg__min_child_weight': 0.02318473553981041, 'reg__feature_fraction': 0.5847409812270346, 'reg__bagging_fraction': 0.6062472698250763, 'reg__bagging_freq': 2, 'reg__max_bin': 149, 'reg__min_data_per_group': 200}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.012468727403158658, 'reg__num_leaves': 32, 'reg__subsample': 0.7693481629765282, 'reg__colsample_bytree': 0.8941817732746274, 'reg__min_child_samples': 50, 'reg__reg_alpha': 0.05342668267123596, 'reg__reg_lambda': 28.811663162928856, 'reg__min_split_gain': 2.3498923163512333, 'reg__path_smooth': 2.872142287173981, 'reg__min_child_weight': 0.0013822306794498897, 'reg__feature_fraction': 0.5435362590420971, 'reg__bagging_fraction': 0.5072958946597673, 'reg__bagging_freq': 2, 'reg__max_bin': 81, 'reg__min_data_per_group': 173}

Rank 6: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.03539518985940601, 'reg__num_leaves': 35, 'reg__subsample': 0.7912585617859196, 'reg__colsample_bytree': 0.6111398317872485, 'reg__min_child_samples': 26, 'reg__reg_alpha': 3.033692030301791, 'reg__reg_lambda': 32.36243965150051, 'reg__min_split_gain': 2.0605275237215714, 'reg__path_smooth': 2.18895823108882, 'reg__min_child_weight':

0.005056450426819884, 'reg__feature_fraction': 0.8064180939878581, 'reg__bagging_fraction': 0.5814865573534707, 'reg__bagging_freq': 4, 'reg__max_bin': 111, 'reg__min_data_per_group': 185}

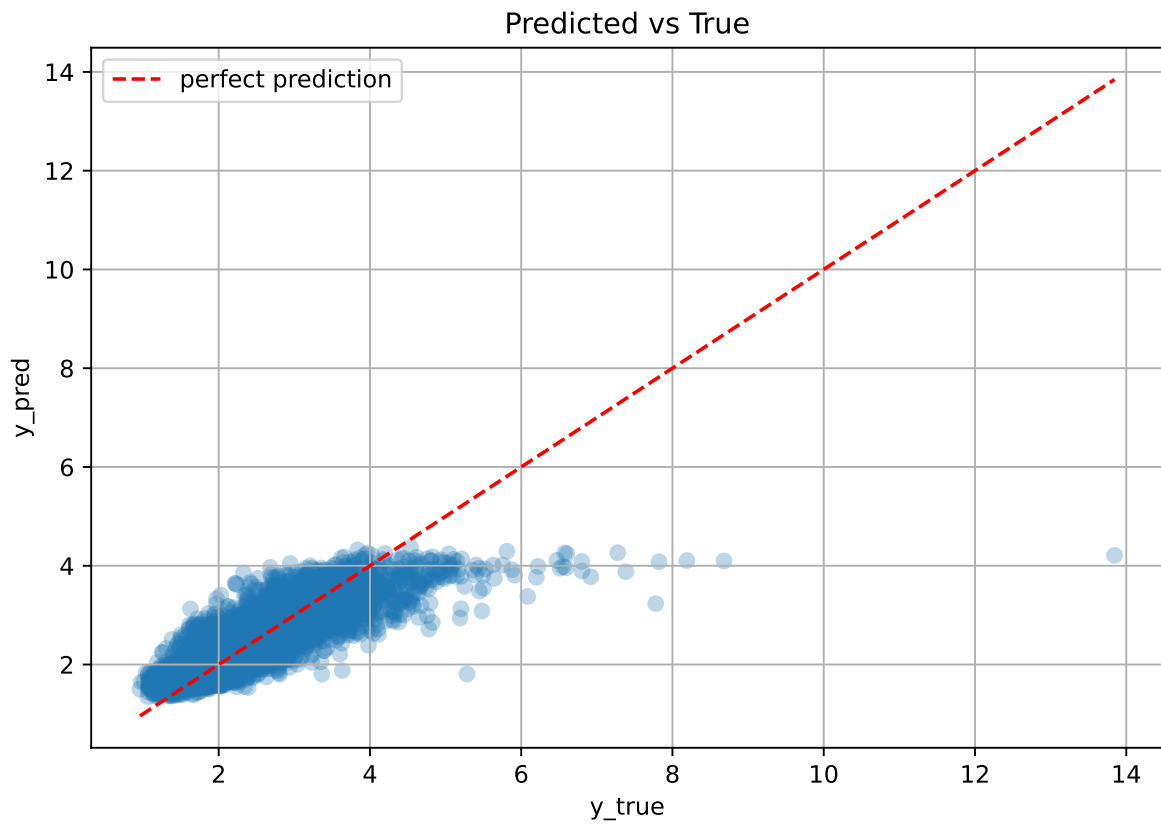
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.020691241278191018, 'reg__num_leaves': 48, 'reg__subsample': 0.7416801862124469, 'reg__colsample_bytree': 0.8462808810503251, 'reg__min_child_samples': 10, 'reg__reg_alpha': 4.7771916554253675, 'reg__reg_lambda': 21.570405433408023, 'reg__min_split_gain': 2.1296635103456243, 'reg__path_smooth': 1.3626942707235, 'reg__min_child_weight': 0.004568187119802103, 'reg__feature_fraction': 0.5768262779922667, 'reg__bagging_fraction': 0.53630760014013, 'reg__bagging_freq': 1, 'reg__max_bin': 91, 'reg__min_data_per_group': 130}

Rank 8: {'reg__n_estimators': 500, 'reg__max_depth': 5, 'reg__learning_rate': 0.021225568336584805, 'reg__num_leaves': 49, 'reg__subsample': 0.7919973986522594, 'reg__colsample_bytree': 0.8618720999494229, 'reg__min_child_samples': 26, 'reg__reg_alpha': 2.8215238799487, 'reg__reg_lambda': 32.59037459215053, 'reg__min_split_gain': 2.0727196333002755, 'reg__path_smooth': 2.455039048113237, 'reg__min_child_weight': 0.0048745127107409, 'reg__feature_fraction': 0.7985567702058051, 'reg__bagging_fraction': 0.5758533406589555, 'reg__bagging_freq': 4, 'reg__max_bin': 113, 'reg__min_data_per_group': 185}

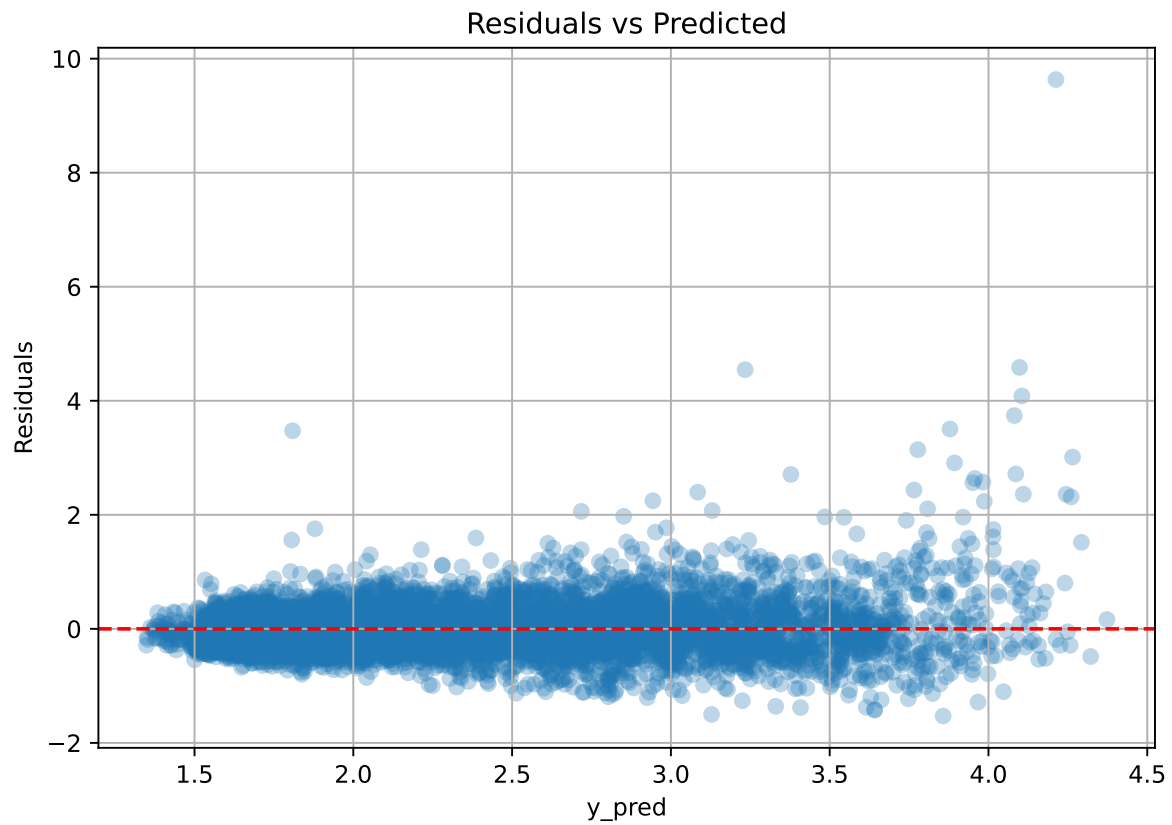
Rank 9: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.03069598833126381, 'reg__num_leaves': 43, 'reg__subsample': 0.7640094281479564, 'reg__colsample_bytree': 0.8980809192804023, 'reg__min_child_samples': 37, 'reg__reg_alpha': 6.0136642817994375, 'reg__reg_lambda': 26.86394020377396, 'reg__min_split_gain': 1.9220763682499644, 'reg__path_smooth': 1.3640318790637331, 'reg__min_child_weight': 0.0015789295557926872, 'reg__feature_fraction': 0.667792634465303, 'reg__bagging_fraction': 0.5645614908660579, 'reg__bagging_freq': 1, 'reg__max_bin': 75, 'reg__min_data_per_group': 129}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.015484268872170965, 'reg__num_leaves': 36, 'reg__subsample': 0.8131845631257447, 'reg__colsample_bytree': 0.6277348161972318, 'reg__min_child_samples': 33, 'reg__reg_alpha': 3.1309617835707995, 'reg__reg_lambda': 29.059503415711795, 'reg__min_split_gain': 2.3420084741004032, 'reg__path_smooth': 1.6574256620435728, 'reg__min_child_weight': 0.0011917022950980378, 'reg__feature_fraction': 0.5573453469795637, 'reg__bagging_fraction': 0.6530556616256009, 'reg__bagging_freq': 2, 'reg__max_bin': 156, 'reg__min_data_per_group': 141}

3.5.2 Scatter



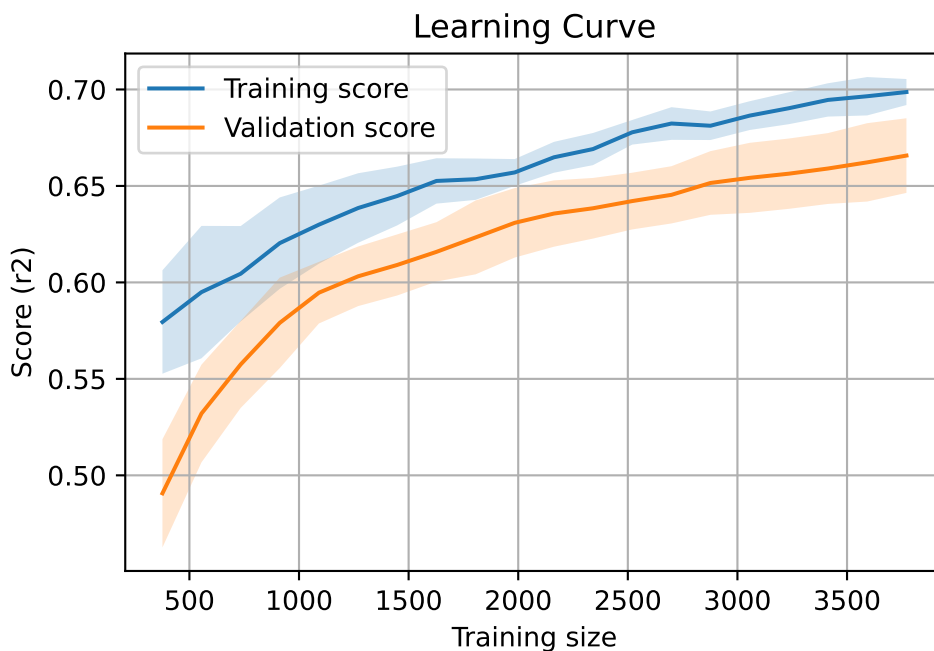
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7698, `Len(training)` : 5389,

`Len(training_real)` : 3773, `Len(test_of_training)` : 1616, `Len(test)` : 2309



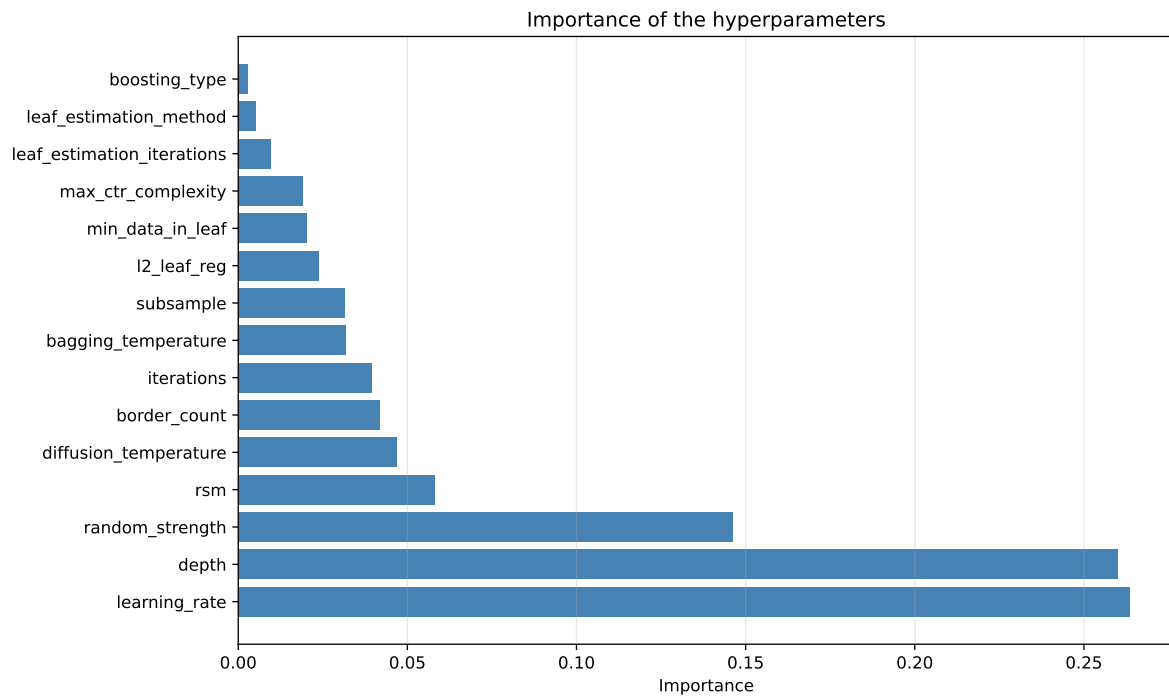
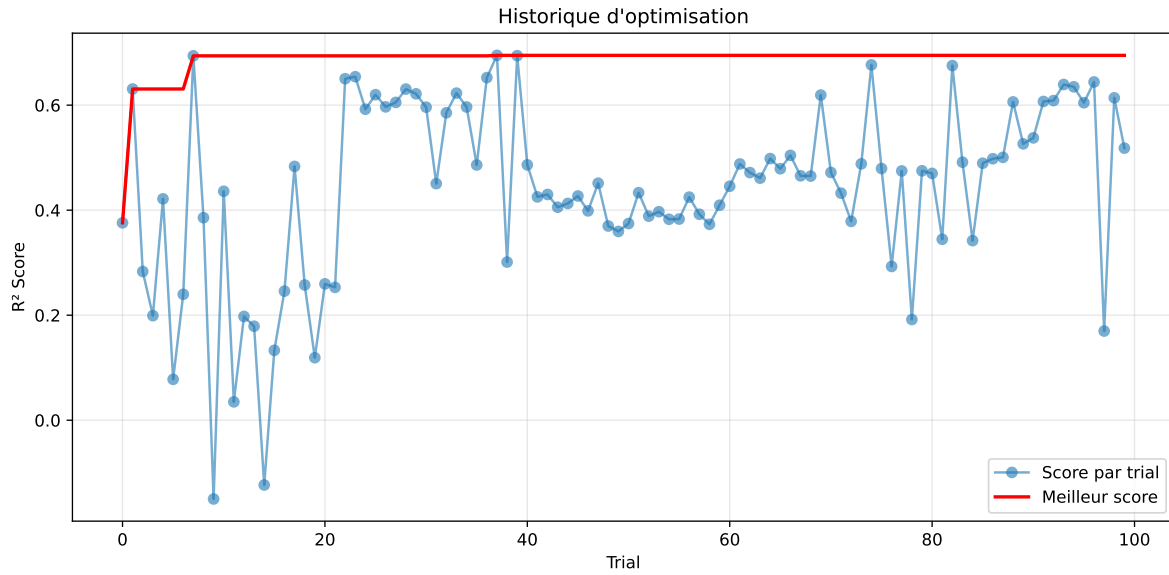
3.6 Model : CatBoost

3.6.1 Gridsearch

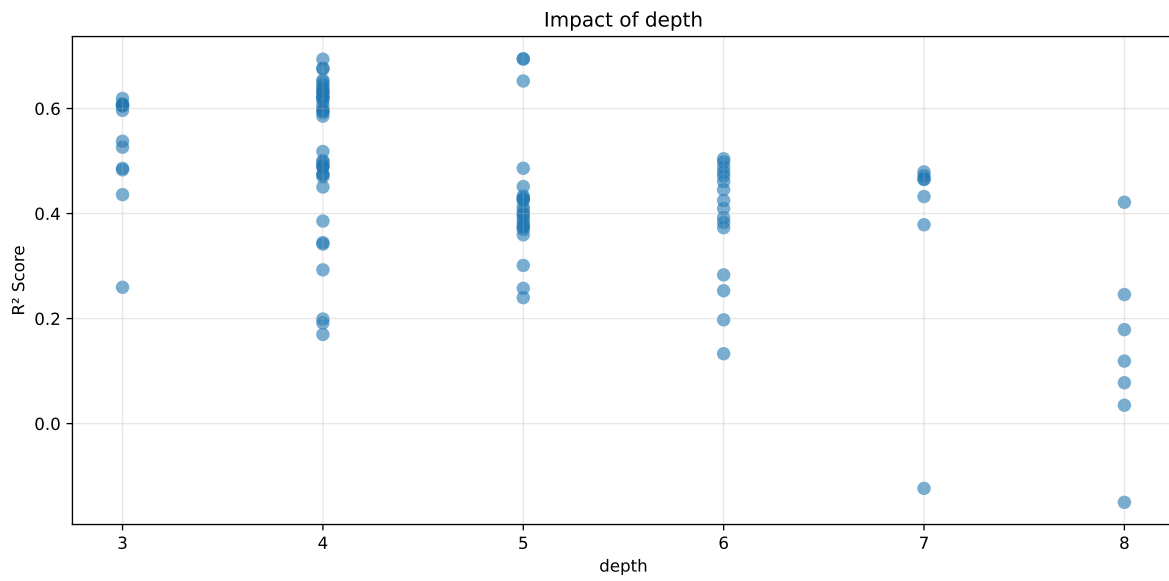
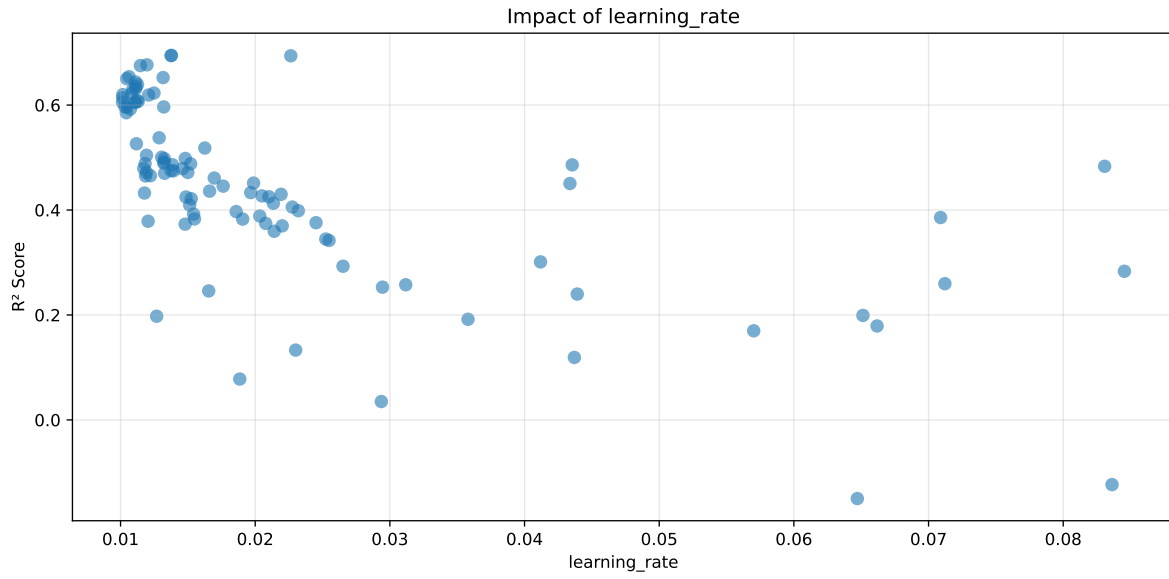
Distribution y_train vs y_test: y_train: mean=2.347, std=0.812, min=0.959, max=13.845
y_test: mean=2.347, std=0.772, min=1.011, max=7.779

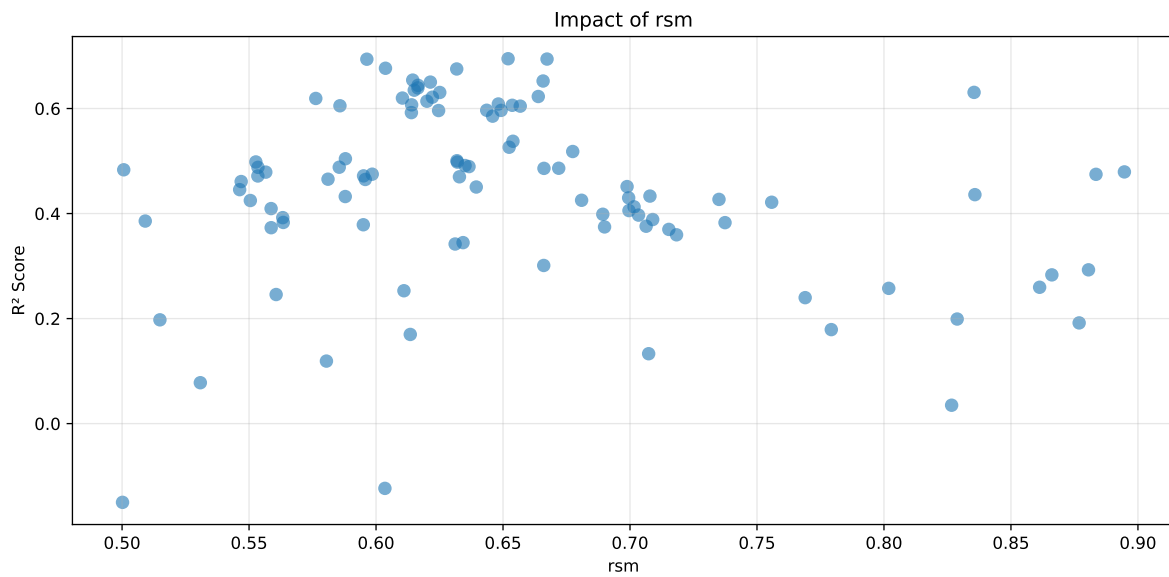
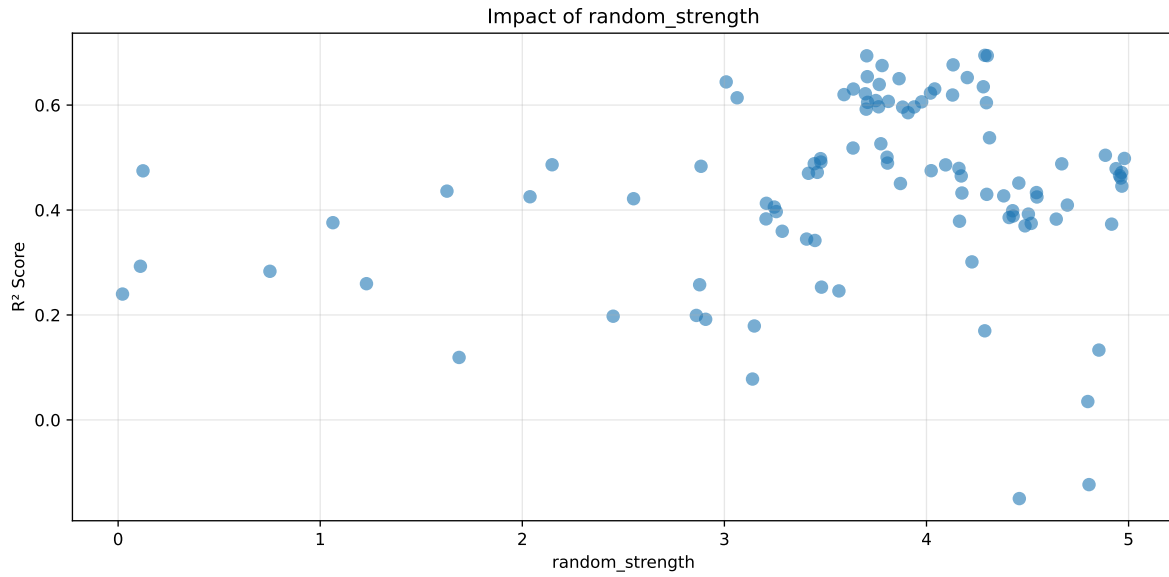
===== TOP Importance FEATURES =====

feature	importance
r_umidity	21.091042
r_temperature_wl_12h_A_energy	15.283437
r_temperature	8.476545
r_temperature_wl_12h_A_energy_rel	6.847613
r_temperature_wl_12h_A_rms	6.319763
r_CO2	5.486579
r_THI	5.216669
r_umidity_wl_12h_A_rms	4.056799
r_NH3_wl_12h_D2_energy	3.144233
r_NH3	3.103227
r_umidity_wl_12h_A_energy	3.102847
r_NH3_wl_12h_A_energy	2.978444
r_NH3_wl_12h_D1_energy	2.789852
r_NH3_wl_12h_D3_energy	2.707379
r_NH3_wl_12h_D3_energy_rel	2.114804
r_NH3_wl_12h_A_rms	1.835372
r_umidity_wl_12h_D3_energy_rel	1.651657
r_NH3_wl_12h_A_energy_rel	1.389372
r_umidity_wl_12h_D1_energy_rel	1.349547
r_temperature_wl_12h_D3_energy_rel	1.054817



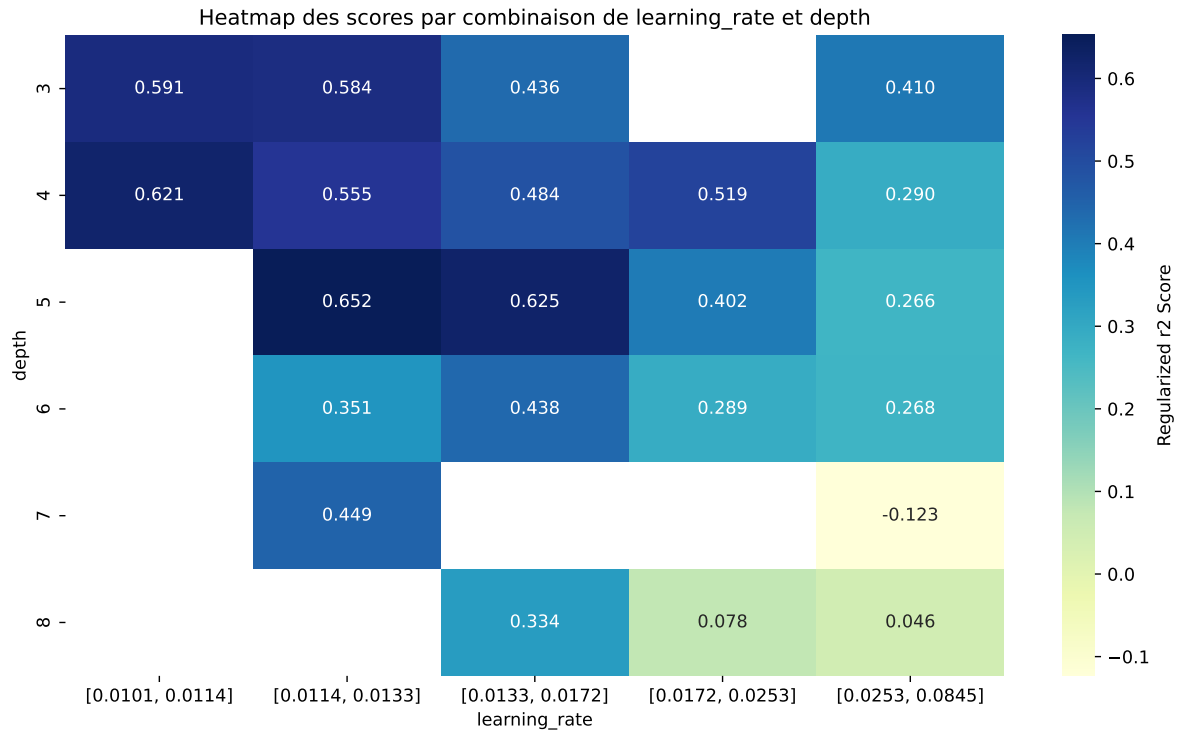
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'random_strength', 'rsm']





C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7288	0.6948	0.7015	0.0196	0.6948	0.7128	0.018	1.0259	1.0489
2	0.7289	0.6942	0.7017	0.0188	0.6942	0.709	0.0148	1.0213	1.05
3	0.7265	0.6939	0.7028	0.0192	0.6939	0.7061	0.0122	1.0176	1.0469
4	0.7081	0.6767	0.6851	0.0182	0.6767	0.6859	0.0092	1.0136	1.0464
5	0.707	0.6753	0.6807	0.0175	0.6753	0.6852	0.0099	1.0146	1.0469
6	0.6754	0.6541	0.6575	0.0163	0.6541	0.6661	0.012	1.0183	1.0326
7	0.675	0.6524	0.6541	0.0152	0.6524	0.6605	0.0082	1.0125	1.0347
8	0.6715	0.6503	0.6552	0.0158	0.6503	0.6638	0.0135	1.0207	1.0326
9	0.6647	0.6441	0.6474	0.0167	0.6441	0.6508	0.0067	1.0104	1.0319
10	0.6587	0.6393	0.6426	0.0162	0.6393	0.6482	0.0089	1.0139	1.0303
---	---	---	---	---	---	---	---	---	---
100	0.9853	0.7961	0.8208	0.0287	-0.1499	0.8336	0.0375	1.0471	1.2377

Hyperparameters:

Rank 1: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.013777017710413397, 'reg__l2_leaf_reg': 44.180105051573086, 'reg__bagging_temperature': 0.01601923606945954, 'reg__random_strength': 4.288790908691465, 'reg__subsample': 0.5766976965824533, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6519977992282593, 'reg__border_count': 73, 'reg__leaf_estimation_method': 'Newton',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 9012.32118637732}

Rank 2: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.013750707697813, 'reg__l2_leaf_reg': 42.70980948038243, 'reg__bagging_temperature': 0.16102309374365342, 'reg__random_strength': 4.300880726565892, 'reg__subsample': 0.6733787259930832, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.6673333106885275, 'reg__border_count': 71, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 3728.9062496378856}

Rank 3: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.022643950927723998, 'reg__l2_leaf_reg': 35.41860483392863, 'reg__bagging_temperature': 3.7363987037935993, 'reg__random_strength': 3.704325257401359, 'reg__subsample': 0.607411326808546, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.5963794141783693, 'reg__border_count': 52, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7590.438677577892}

Rank 4: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.011960583364330262, 'reg__l2_leaf_reg': 38.428745826651316, 'reg__bagging_temperature': 0.32982856429292495, 'reg__random_strength': 4.131624934008541, 'reg__subsample': 0.6565295796605947, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.6037158151745481, 'reg__border_count': 78, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 4599.813320730297}

Rank 5: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.011473259720977859, 'reg__l2_leaf_reg': 36.75306428213835, 'reg__bagging_temperature': 0.5713296608137597, 'reg__random_strength': 3.7802306672358035, 'reg__subsample': 0.5543987246522916, 'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6317698453084992, 'reg__border_count': 63, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 8414.700094867536}

Rank 6: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.010628065573368125, 'reg__l2_leaf_reg': 33.3872004110053, 'reg__bagging_temperature': 0.1349475510185667, 'reg__random_strength': 3.7071438072582734, 'reg__subsample': 0.6433986340446223, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.6144419539923474, 'reg__border_count': 127, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 3515.596955918453}

Rank 7: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.013159885404692383, 'reg__l2_leaf_reg': 22.71941965404499, 'reg__bagging_temperature': 1.0523602030461823, 'reg__random_strength': 4.201797163459536, 'reg__subsample': 0.8914454127370682,

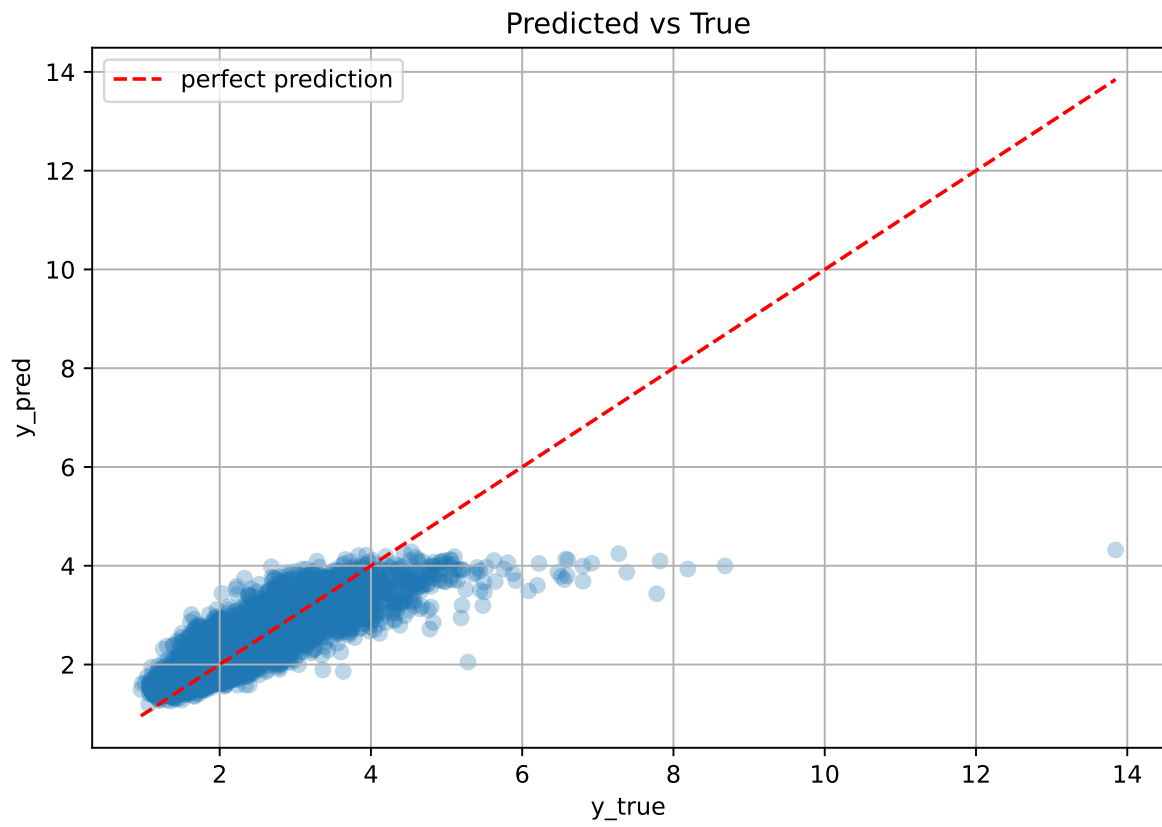
'reg__min_data_in_leaf': 24, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.6657795828637562, 'reg__border_count': 73, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 3663.3170444815246}

Rank 8: {'reg__iterations': 700, 'reg__depth': 4, 'reg__learning_rate': 0.01045854616661267, 'reg__l2_leaf_reg': 35.89587768270384, 'reg__bagging_temperature': 0.17675924175954805, 'reg__random_strength': 3.8642882411878063, 'reg__subsample': 0.651329244581219, 'reg__min_data_in_leaf': 19, 'reg__leaf_estimation_iterations': 9, 'reg__rsm': 0.6213909687903673, 'reg__border_count': 75, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9828.674930972098}

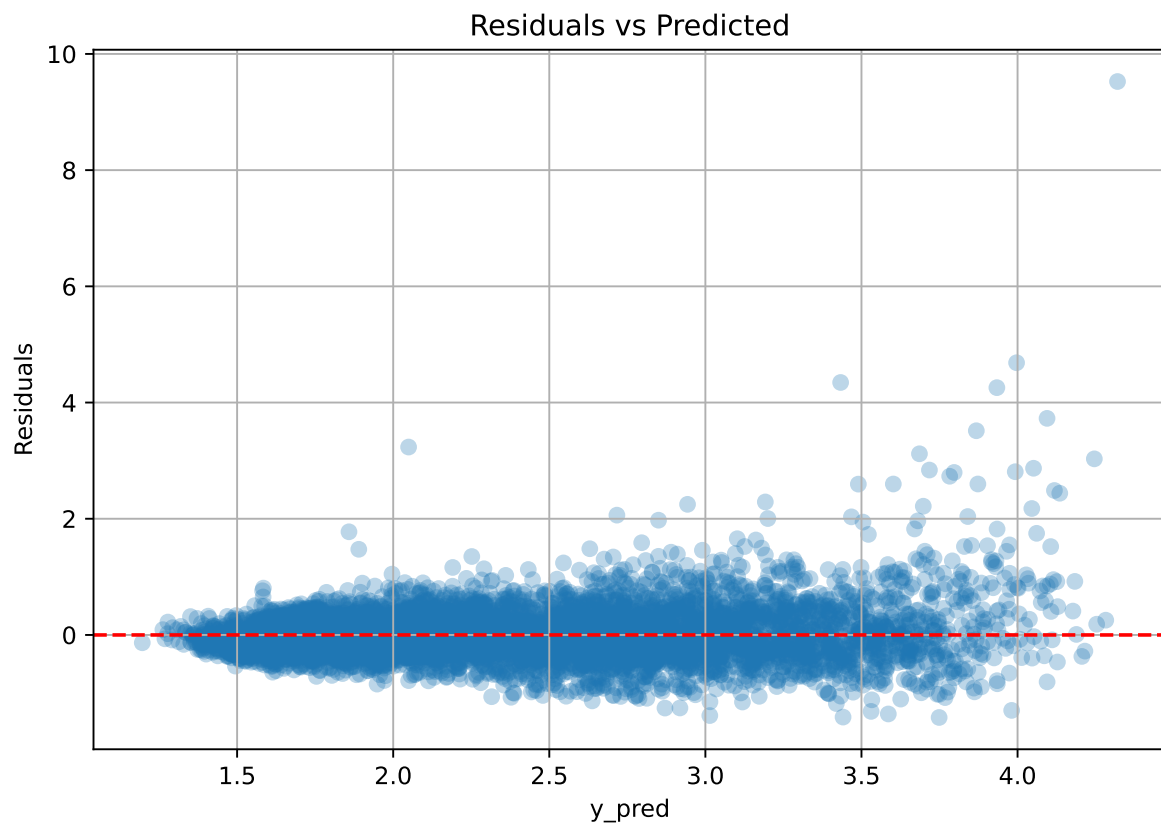
Rank 9: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.011108538935283473, 'reg__l2_leaf_reg': 33.58169097738518, 'reg__bagging_temperature': 0.19670717336321167, 'reg__random_strength': 3.0089071834974623, 'reg__subsample': 0.591312679048638, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.6165397424252765, 'reg__border_count': 77, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5722.940544057745}

Rank 10: {'reg__iterations': 500, 'reg__depth': 4, 'reg__learning_rate': 0.011262403497114552, 'reg__l2_leaf_reg': 33.215617055327456, 'reg__bagging_temperature': 0.1719856510342005, 'reg__random_strength': 3.7662403256840076, 'reg__subsample': 0.5902254092132683, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.6164882594788645, 'reg__border_count': 66, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 9207.242671283226}

3.6.2 Scatter



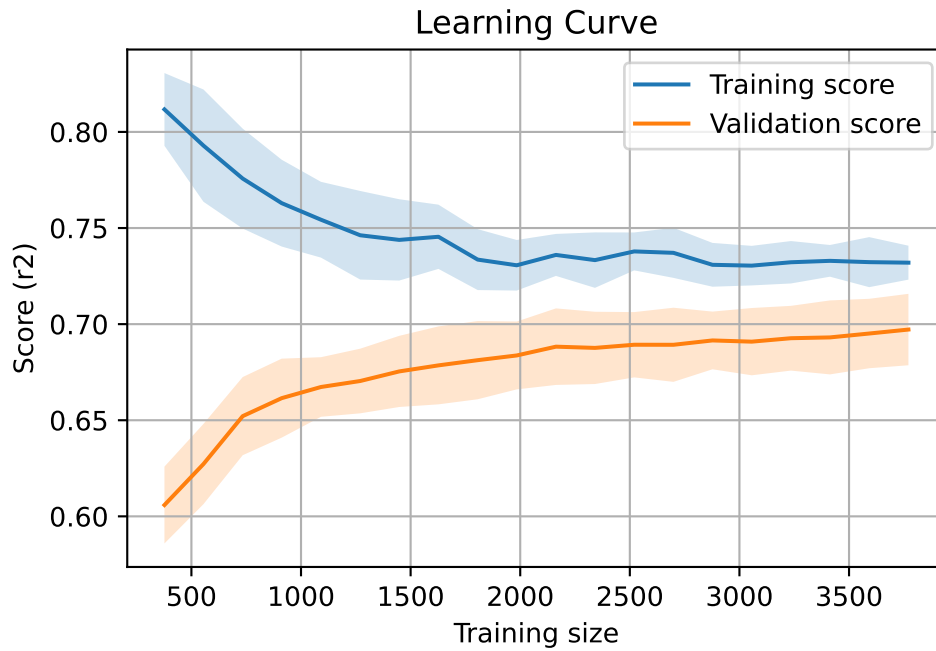
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7698, `Len(training)` : 5389,

`Len(training_real)` : 3773, `Len(test_of_training)` : 1616, `Len(test)` : 2309



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.514	0.504	0.004	0.508	0.509	0.004	1.009	1.02	nan
LGBM	0.697	0.666	0.019	0.671	0.693	0.027	1.041	1.047	0.6656
CatBoost	0.729	0.695	0.02	0.702	0.713	0.018	1.026	1.049	0.6948

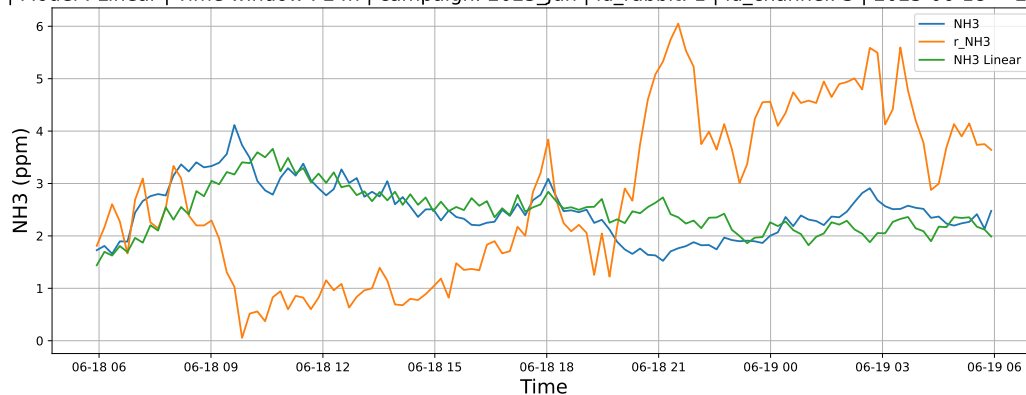
4 Plot

4.1 Standalone plots

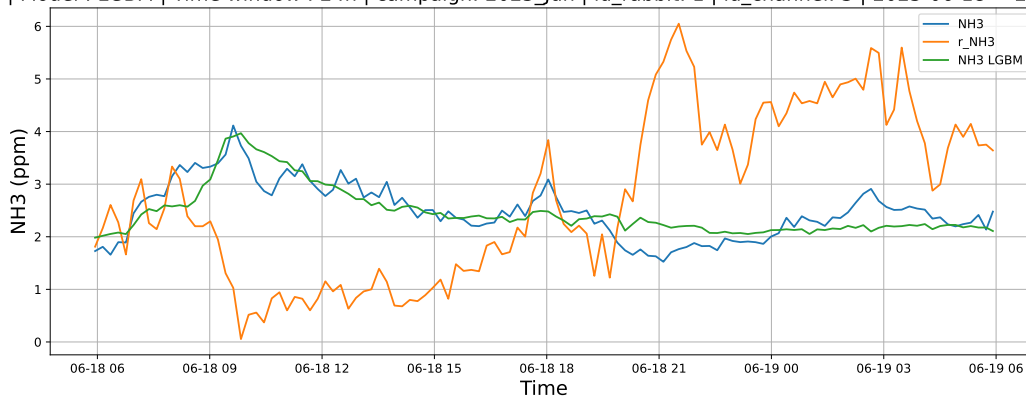
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

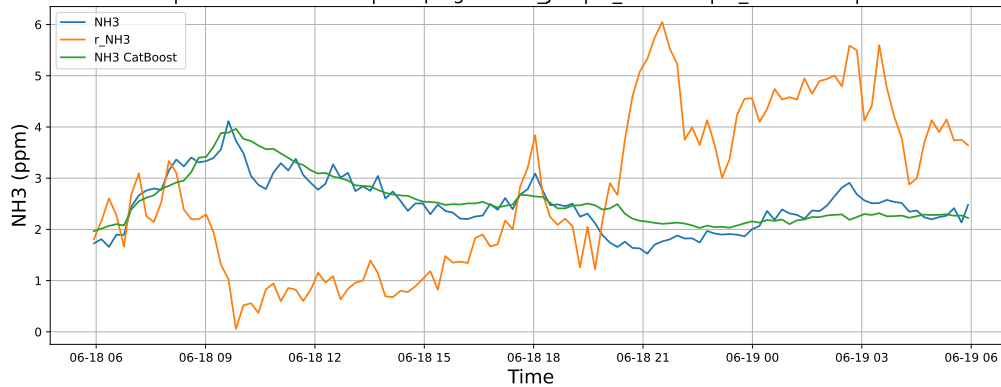
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

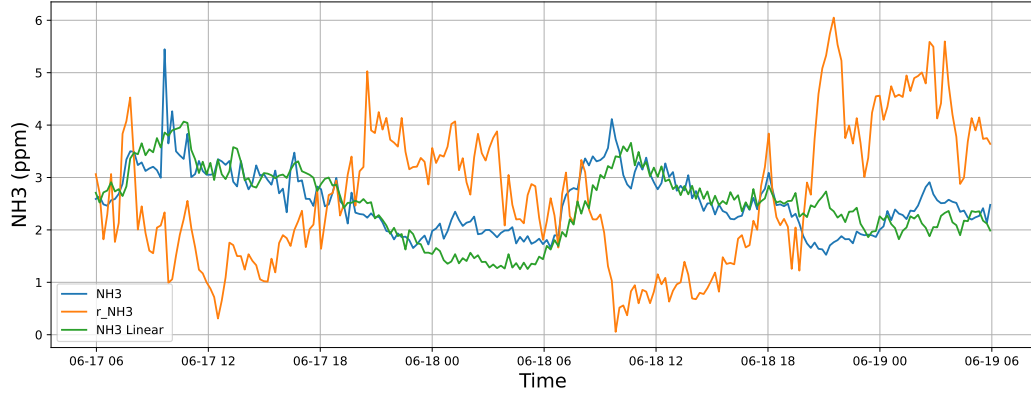


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

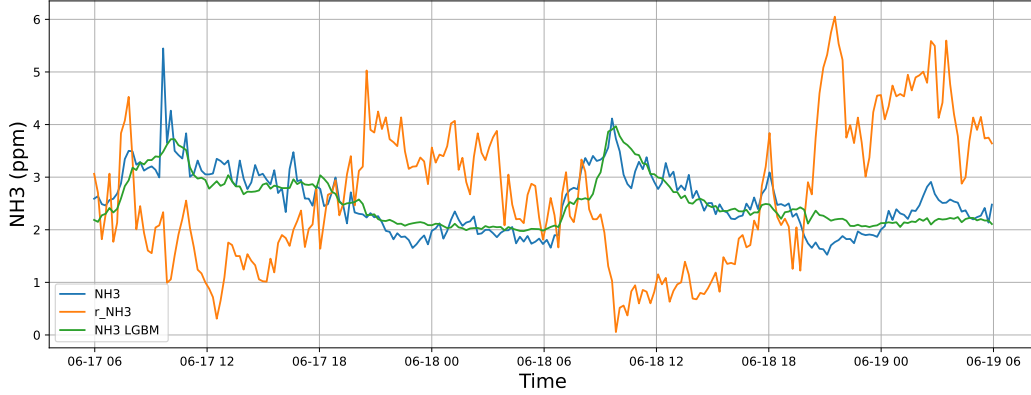


4.1.1.2 Time window : 48

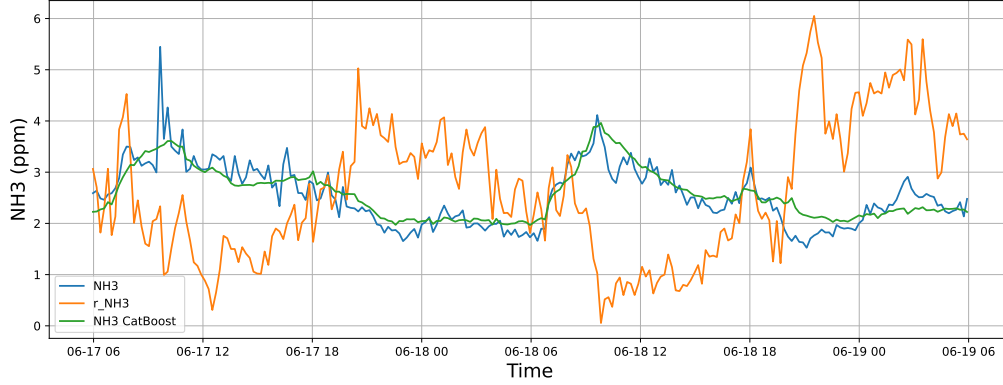
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

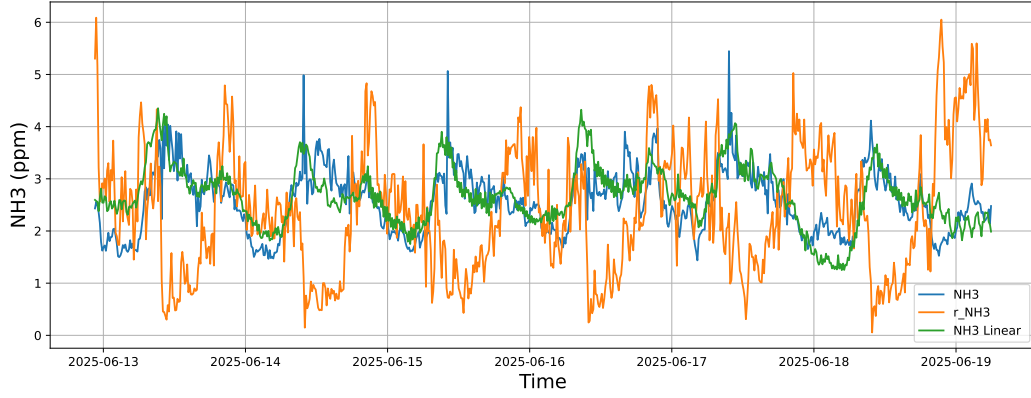


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

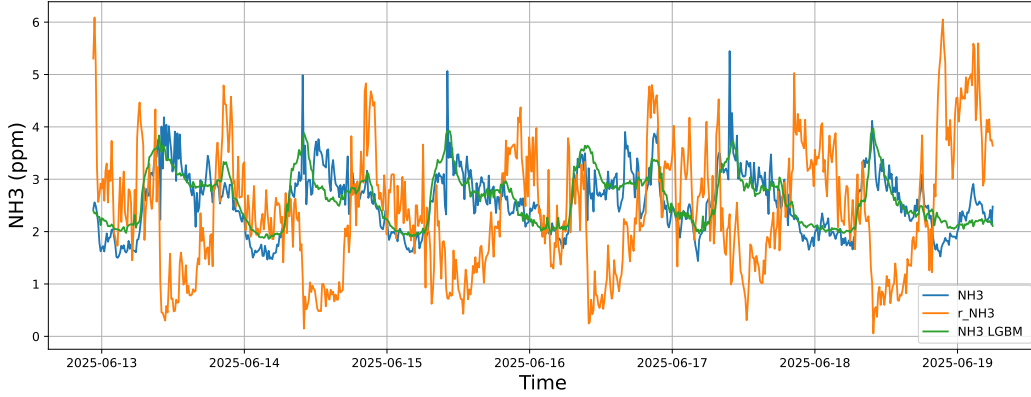


4.1.1.3 Time window : ALL

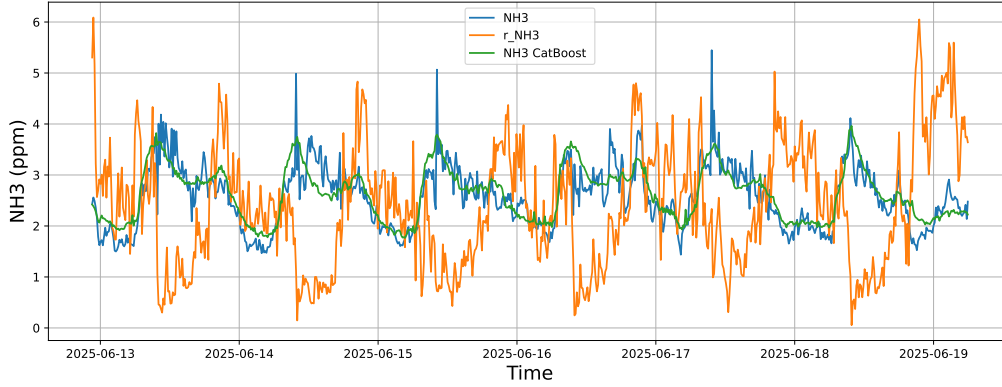
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



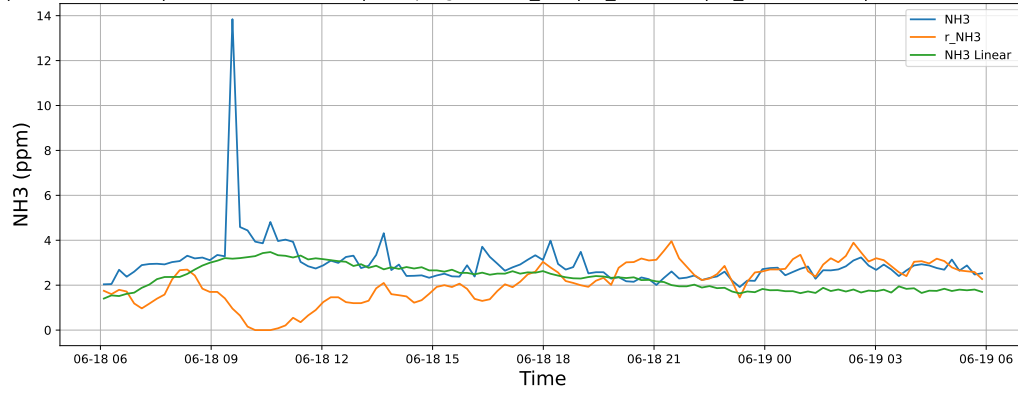
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



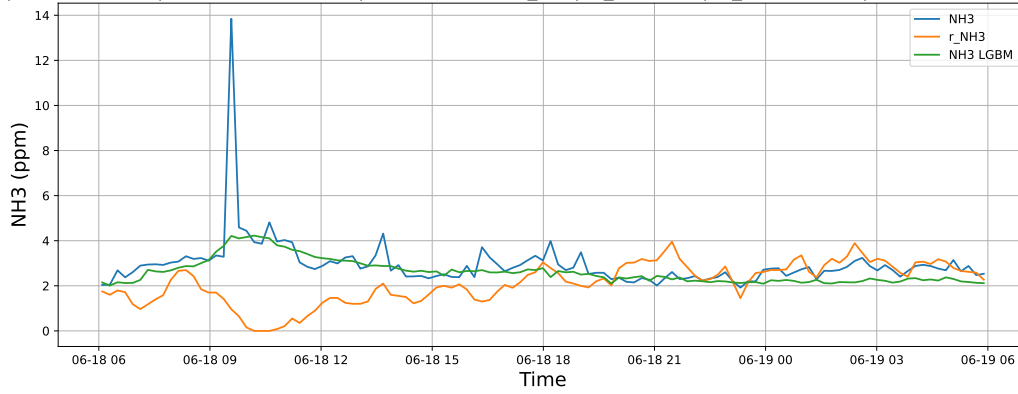
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

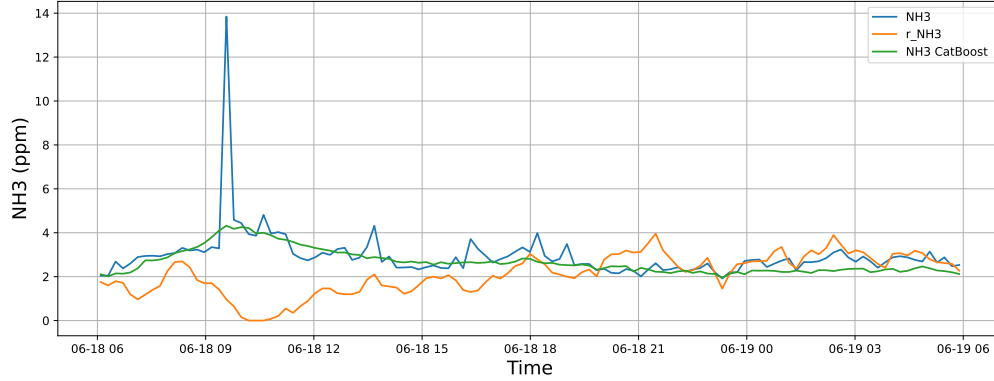
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

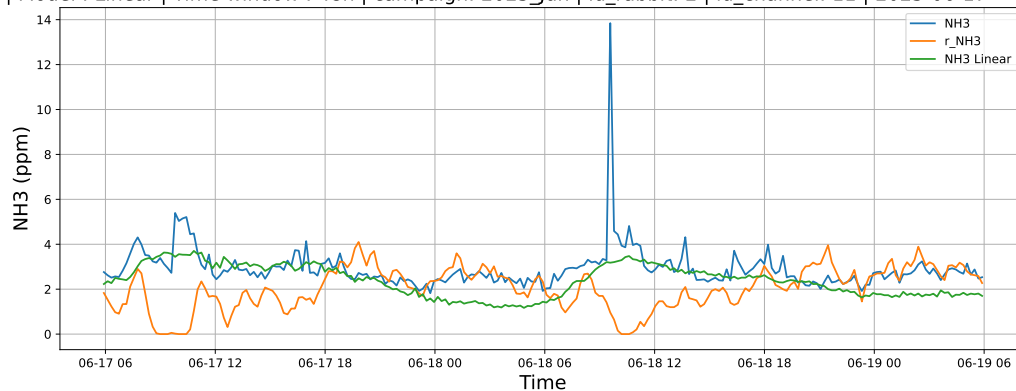


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

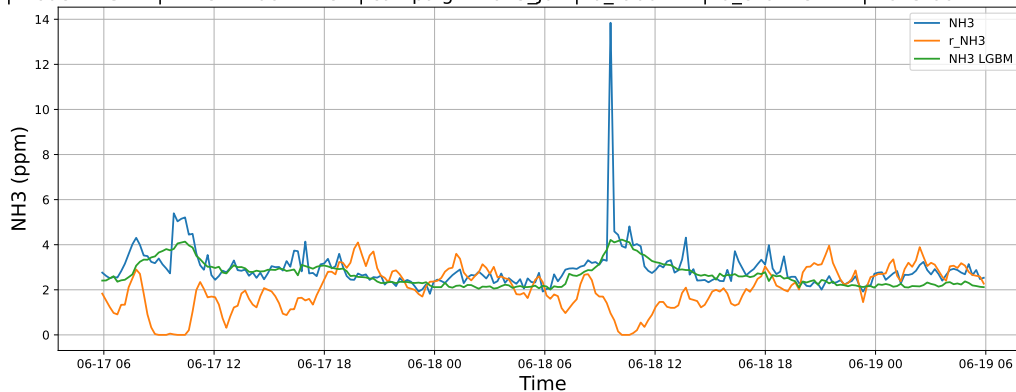


4.1.2.2 Time window : 48

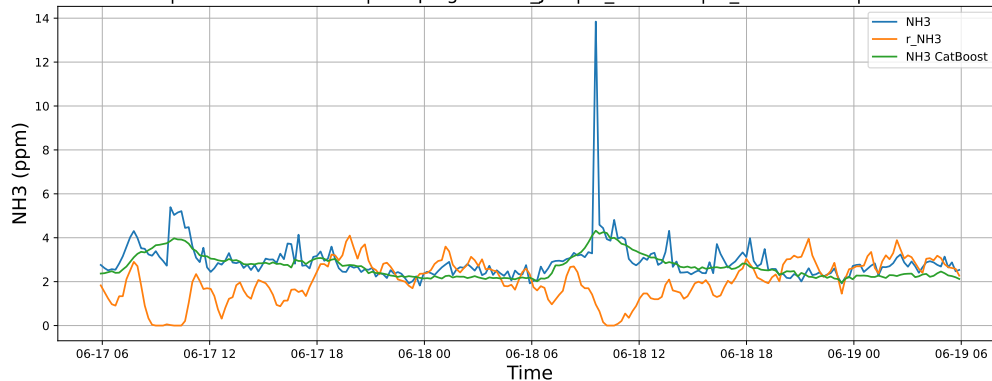
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

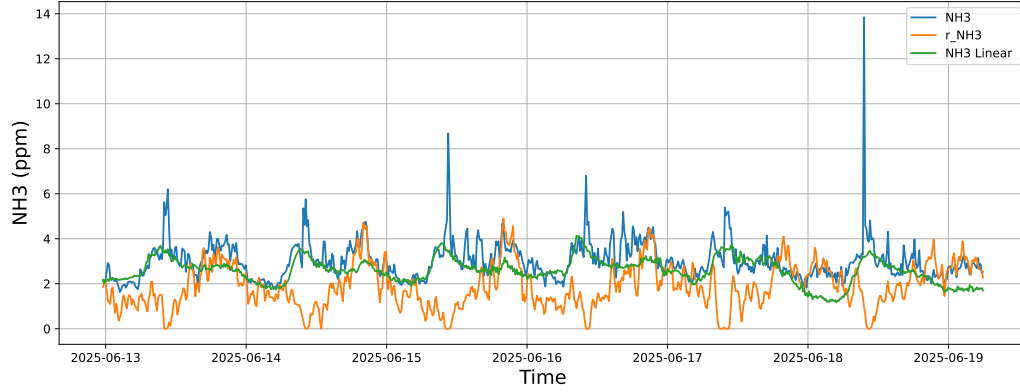


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

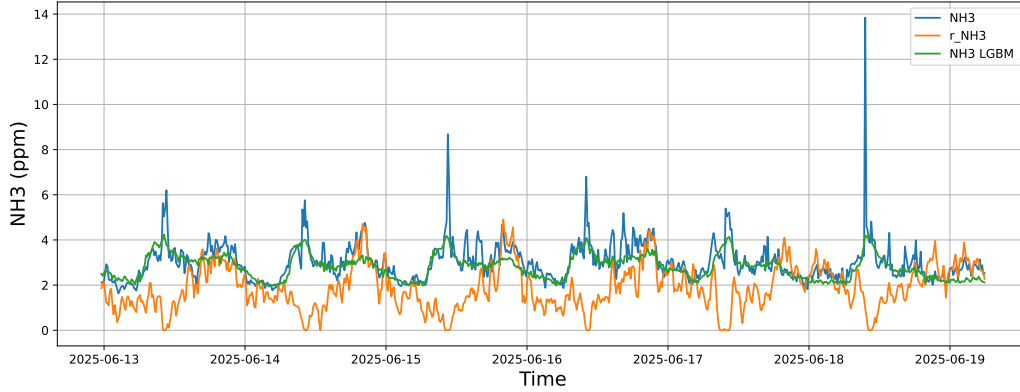


4.1.2.3 Time window : ALL

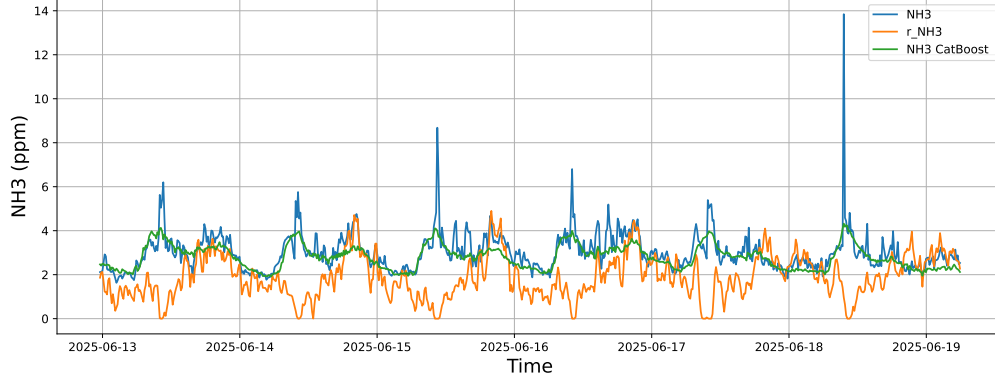
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



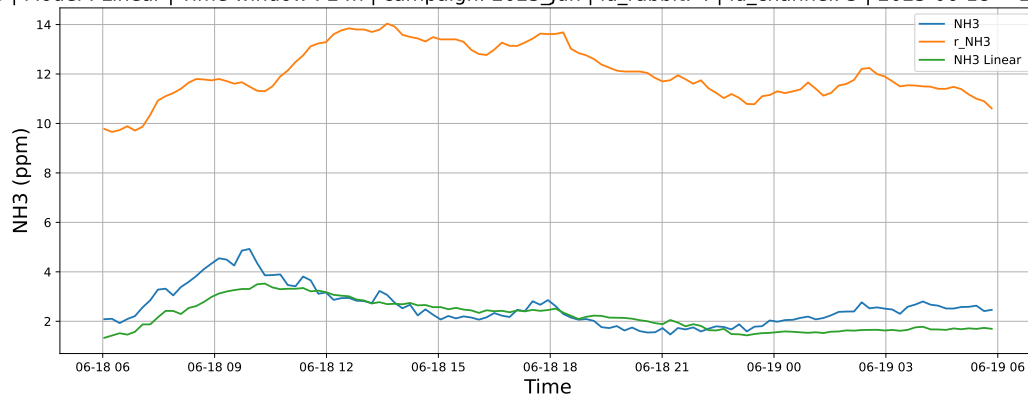
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



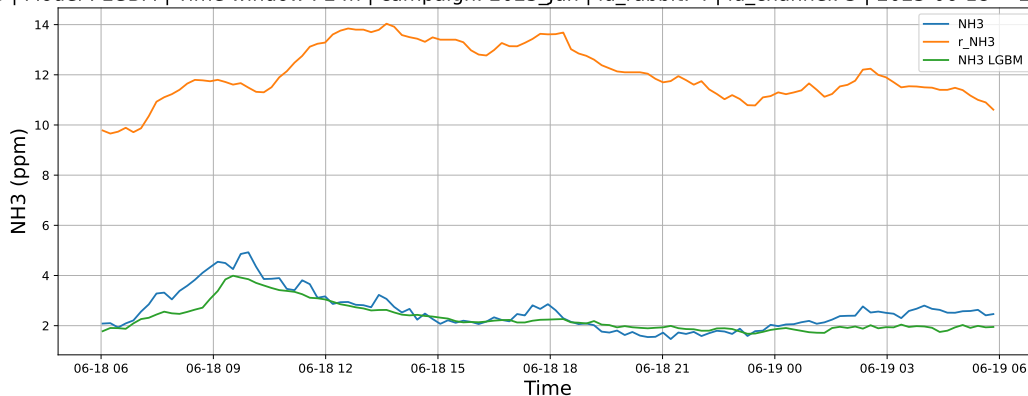
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

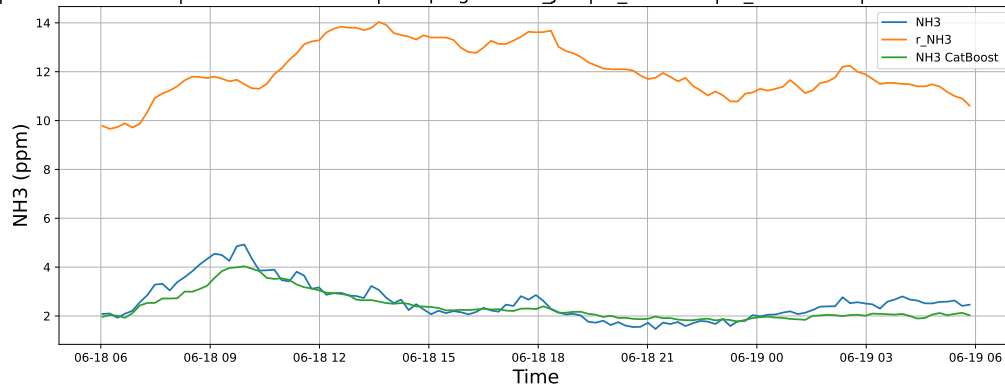
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

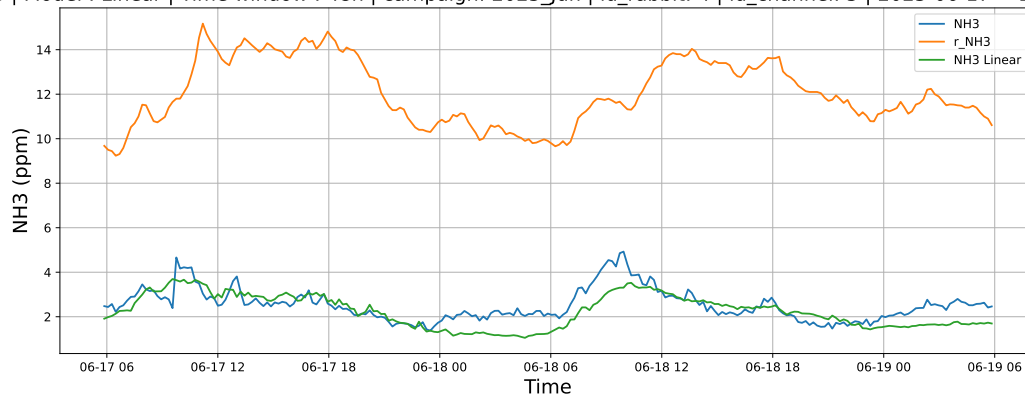


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

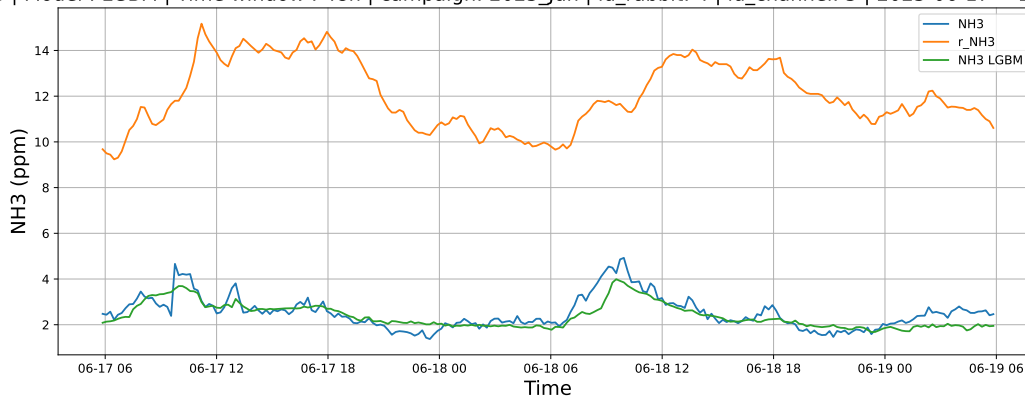


4.1.3.2 Time window : 48

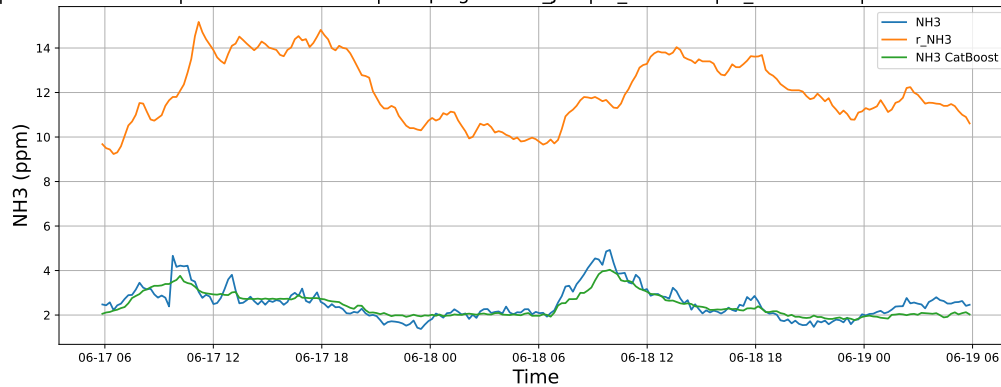
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

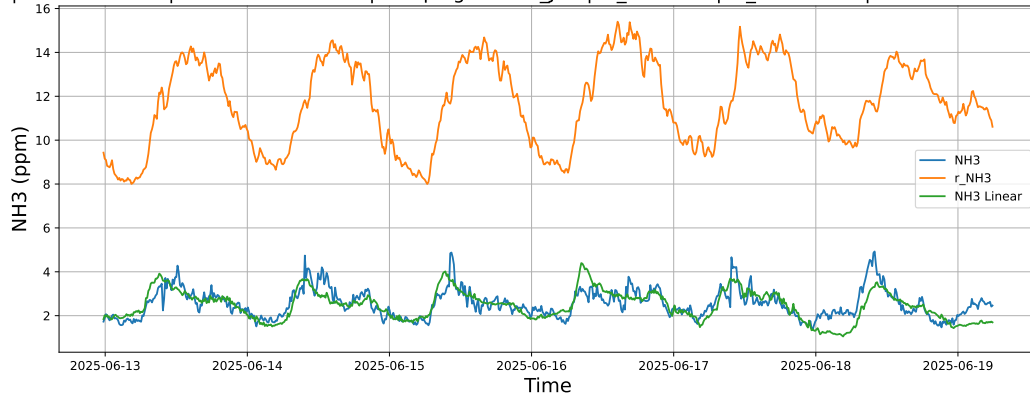


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

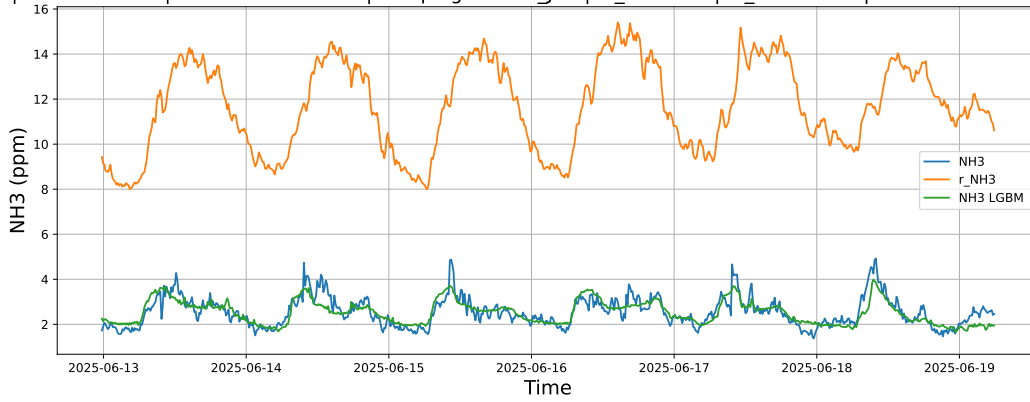


4.1.3.3 Time window : ALL

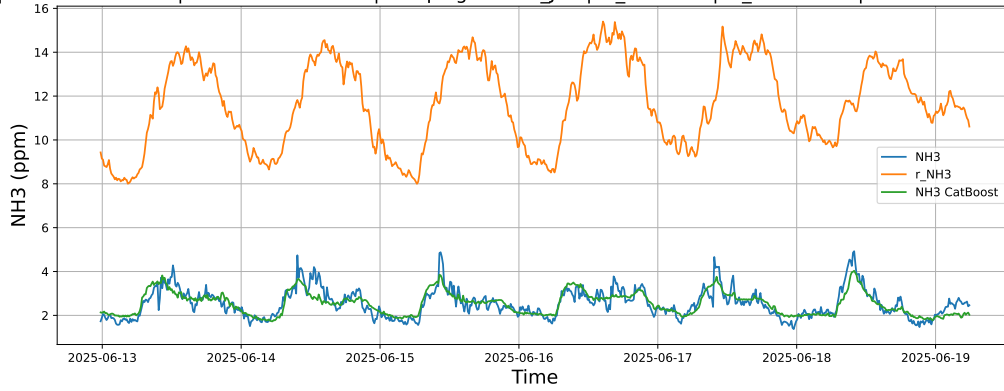
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



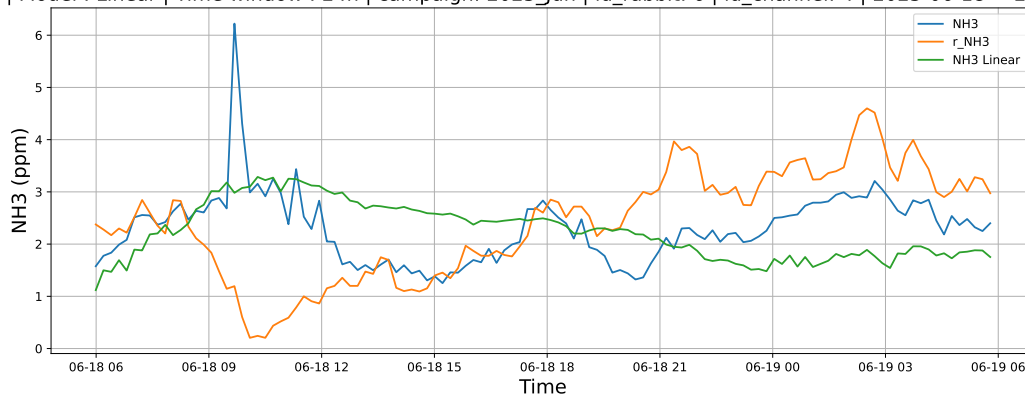
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



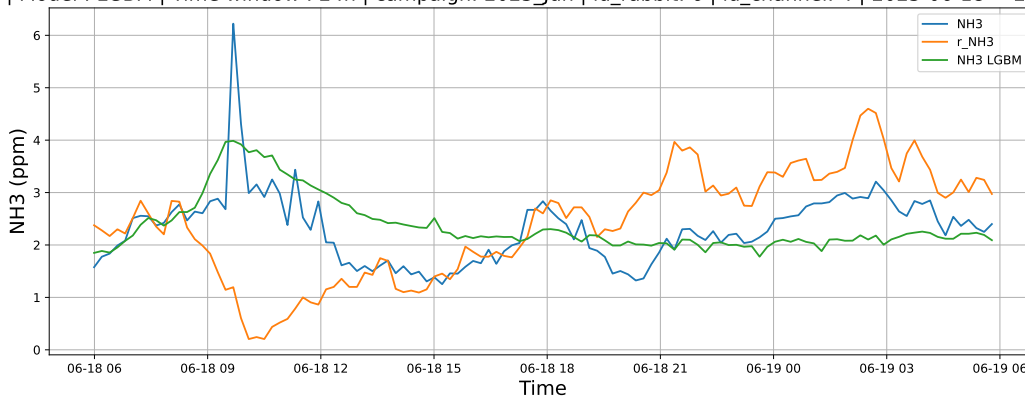
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

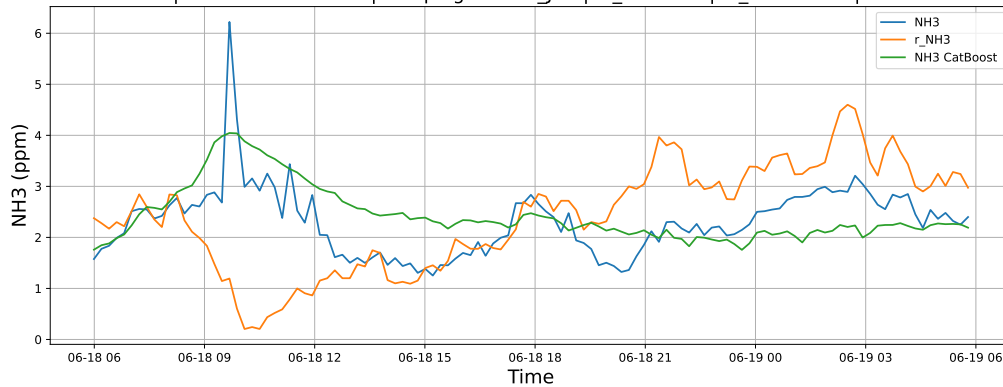
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

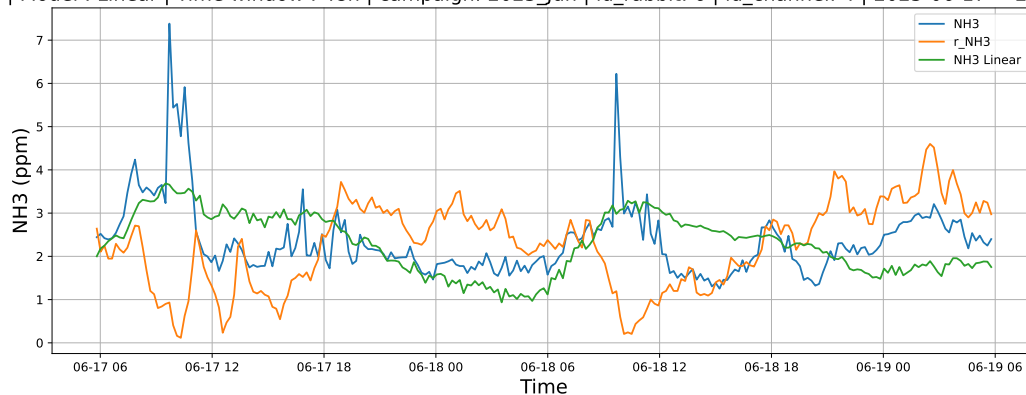


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

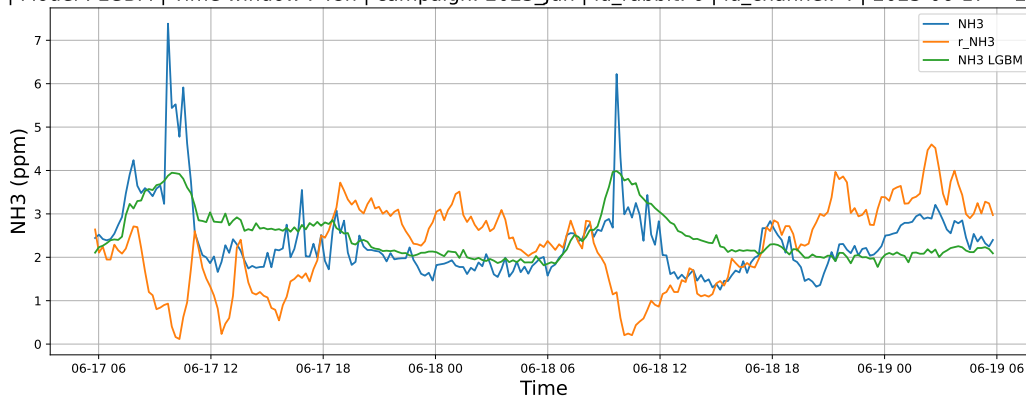


4.1.4.2 Time window : 48

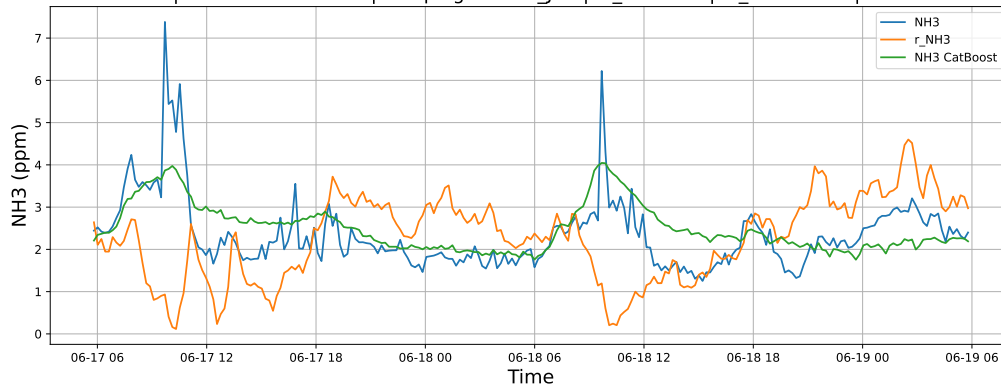
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

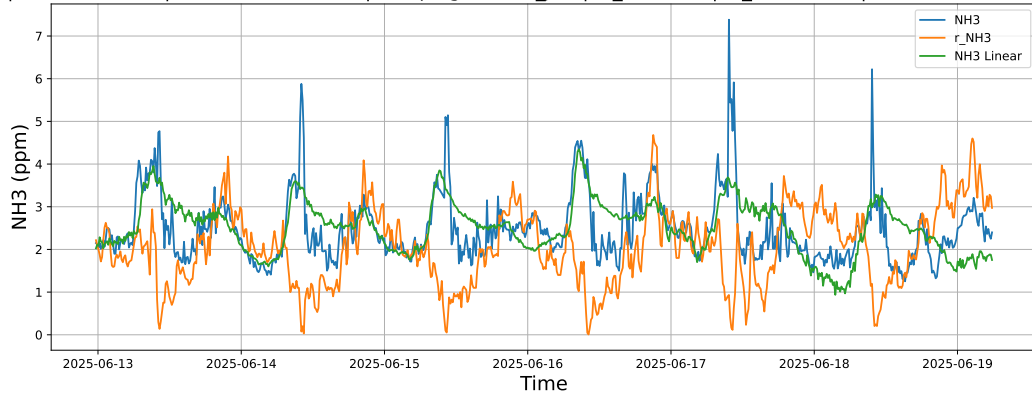


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

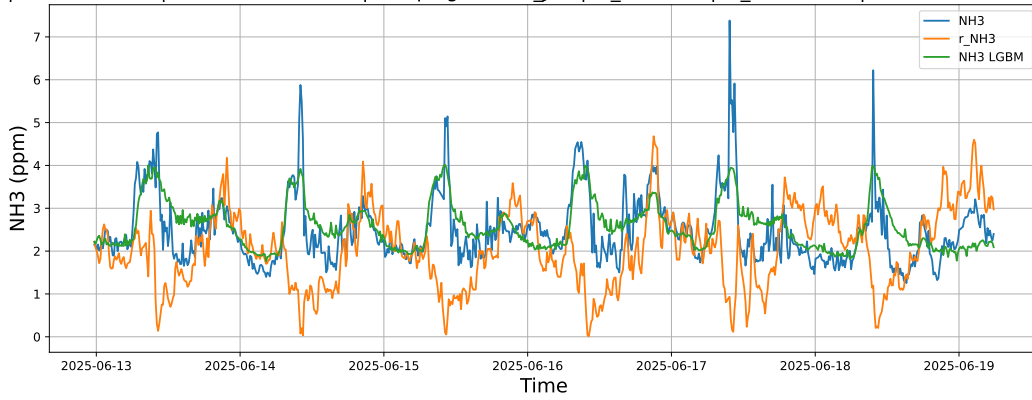


4.1.4.3 Time window : ALL

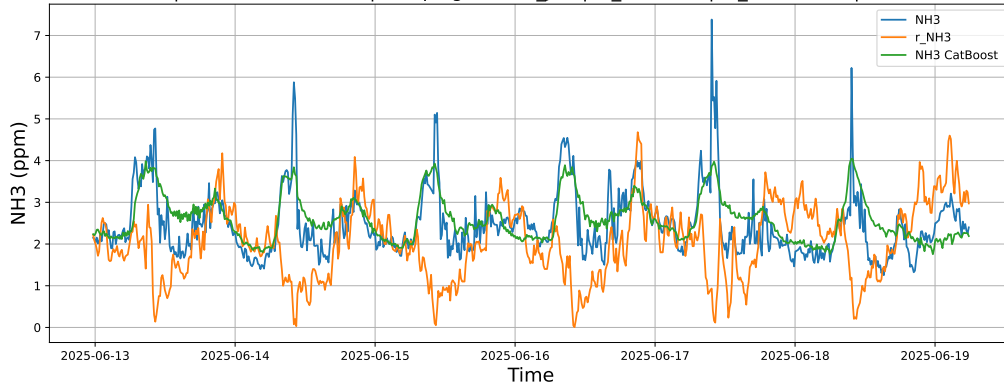
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



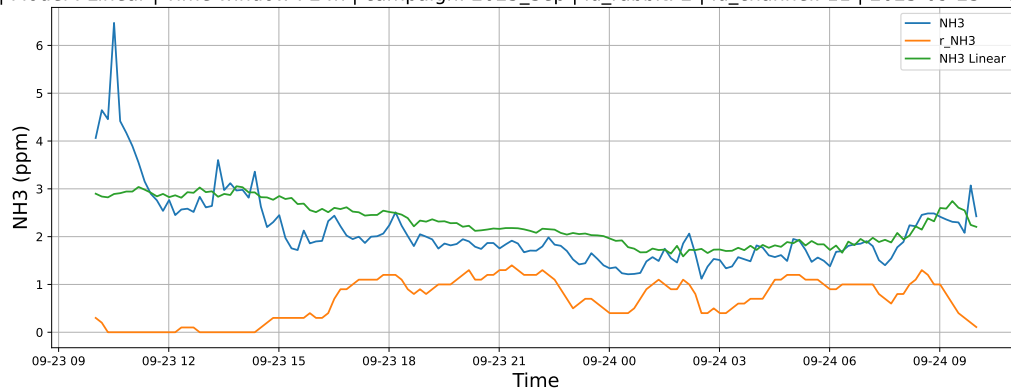
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



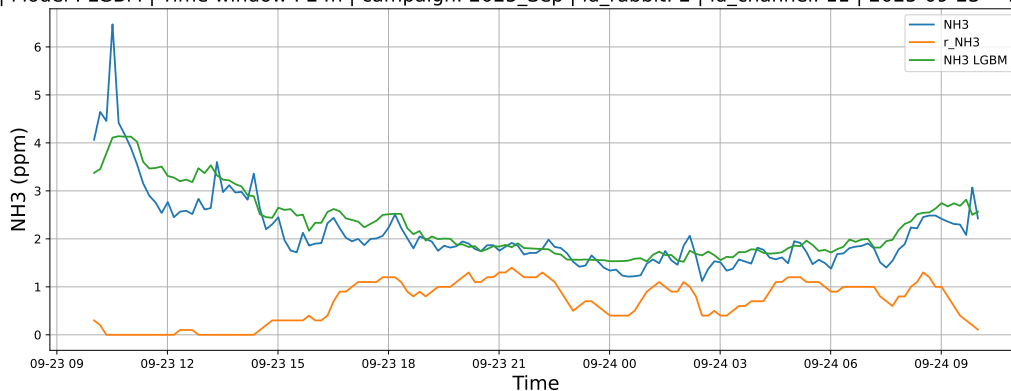
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

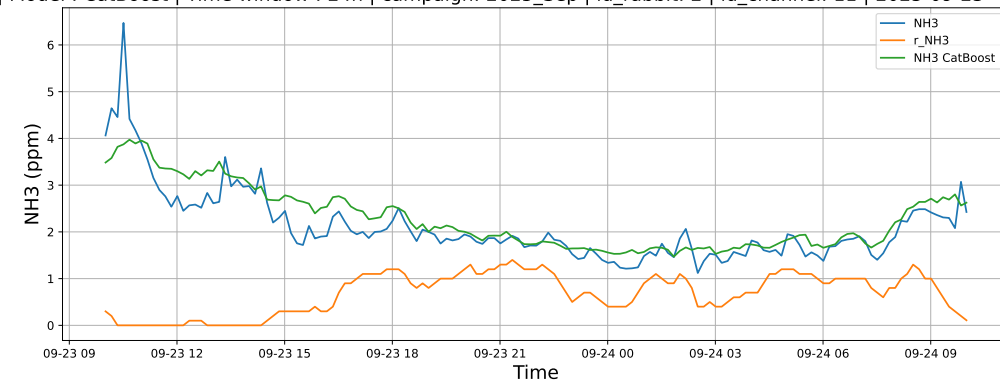
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

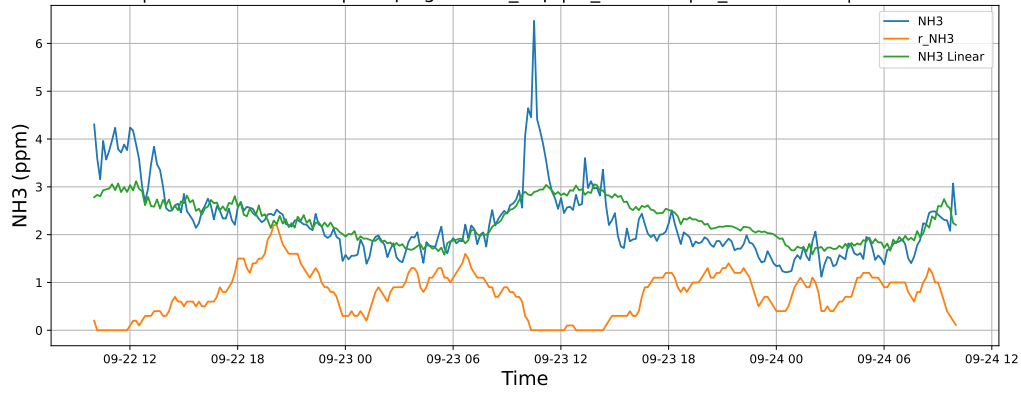


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

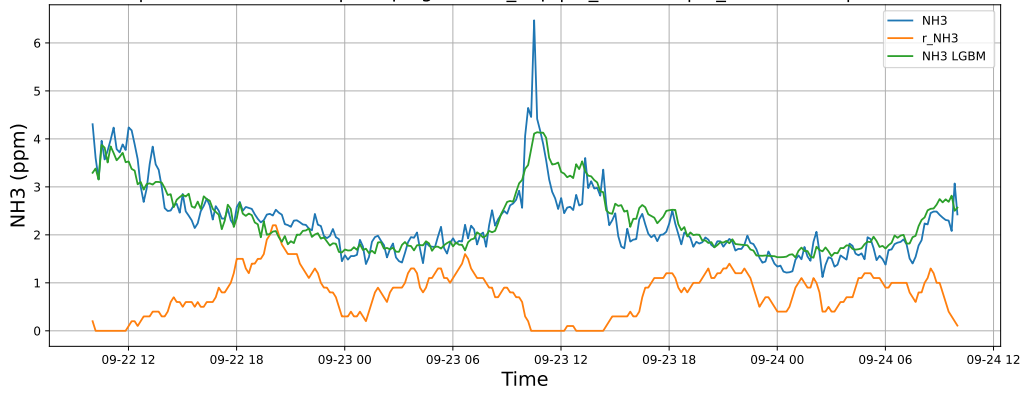


4.1.5.2 Time window : 48

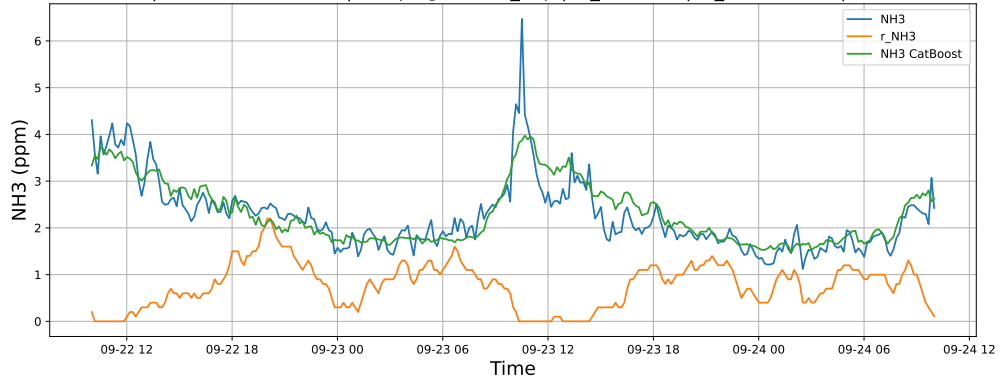
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

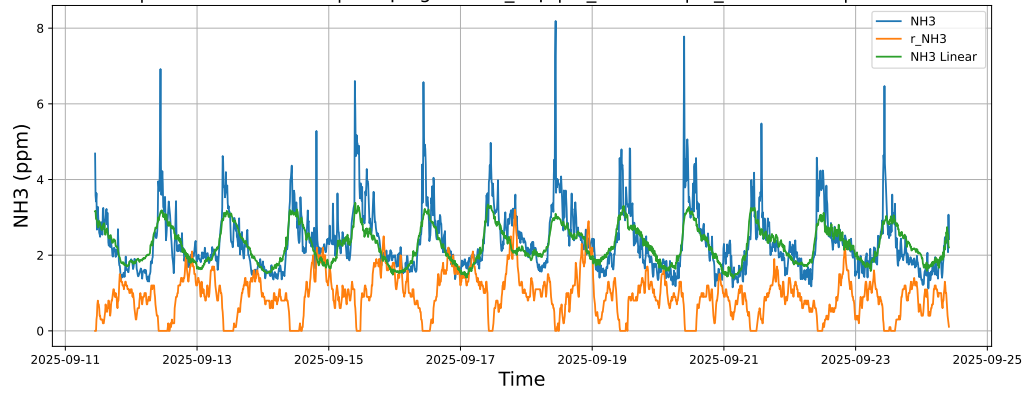


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

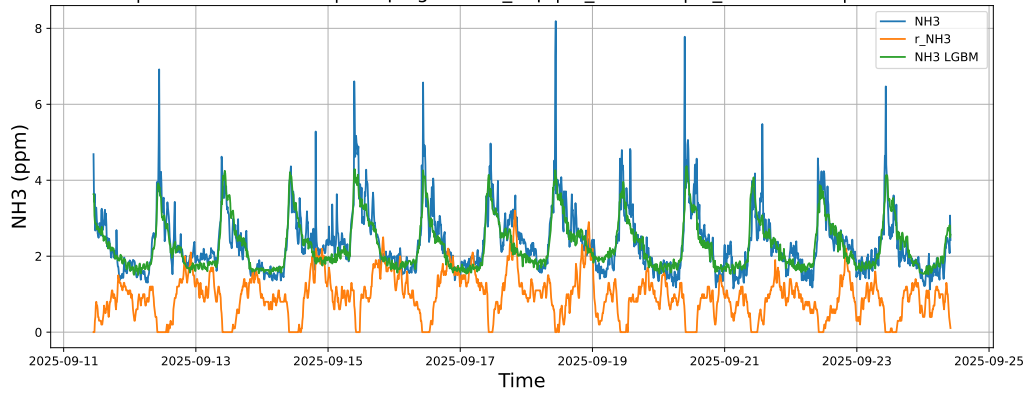


4.1.5.3 Time window : ALL

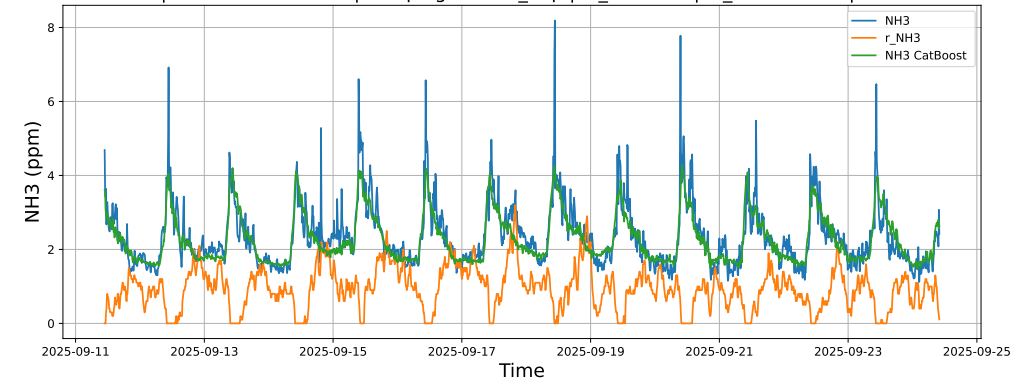
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



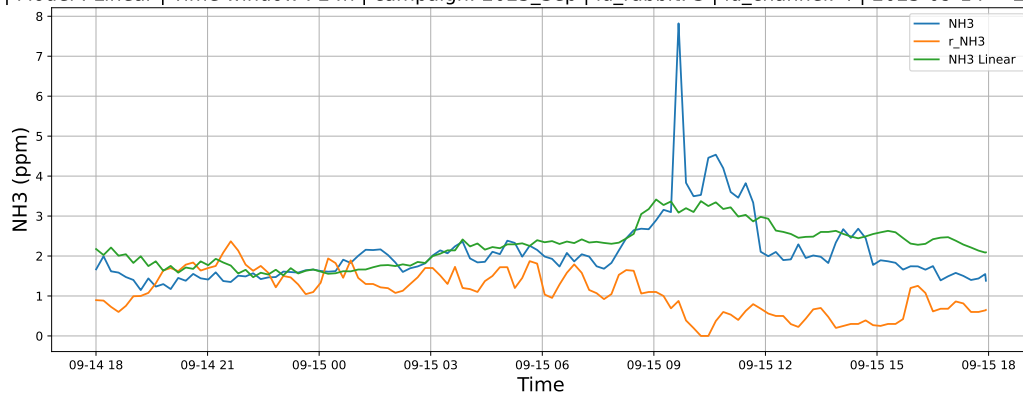
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



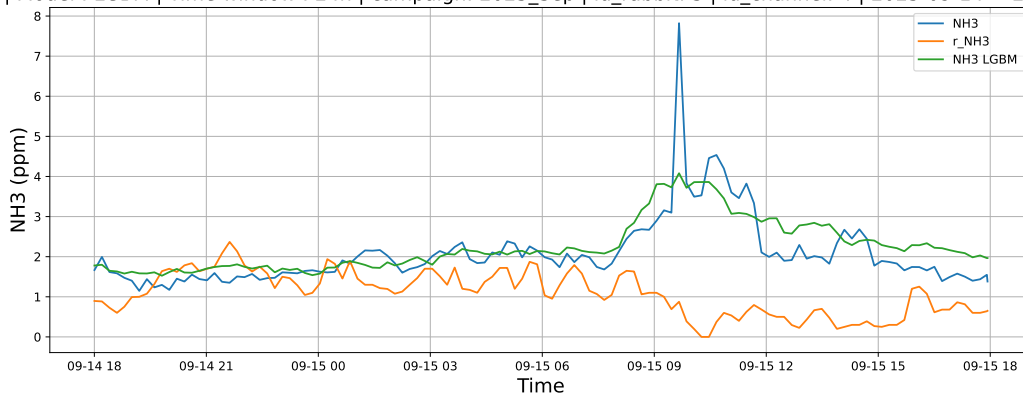
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

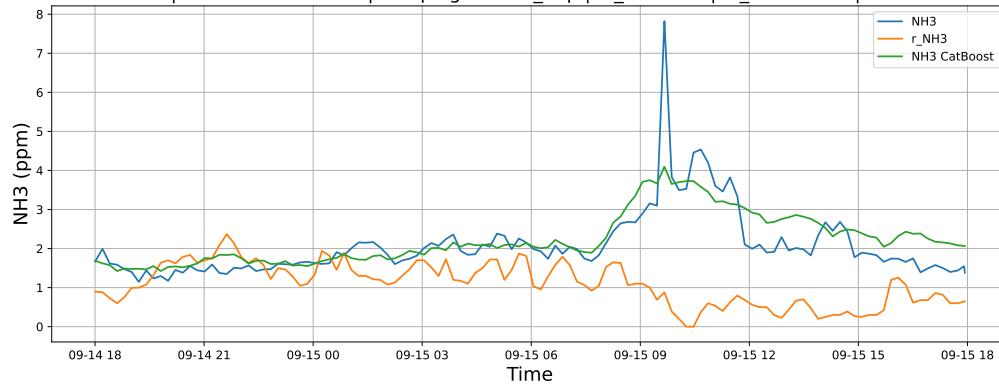
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

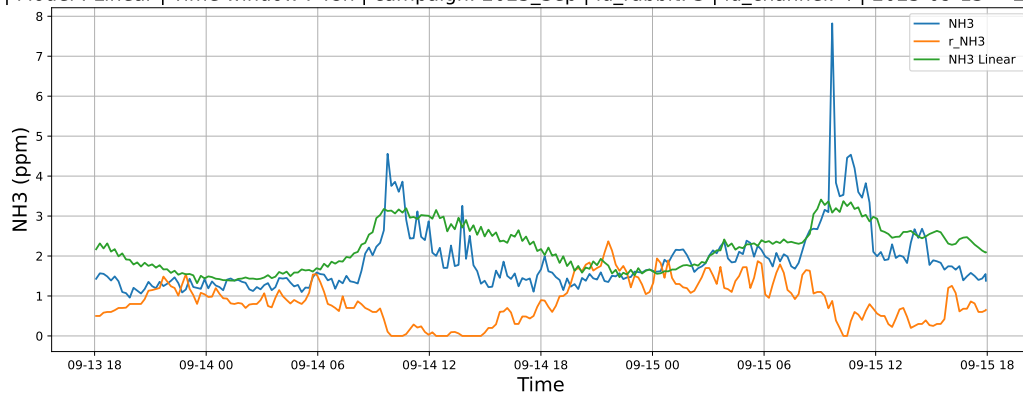


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

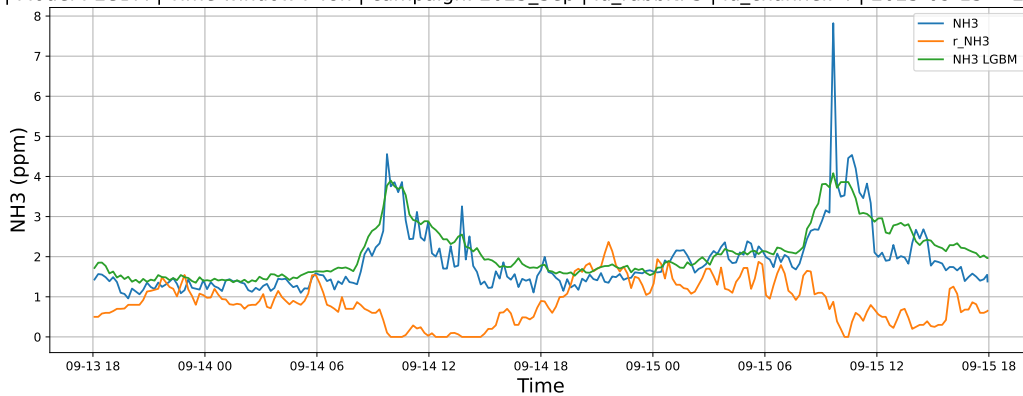


4.1.6.2 Time window : 48

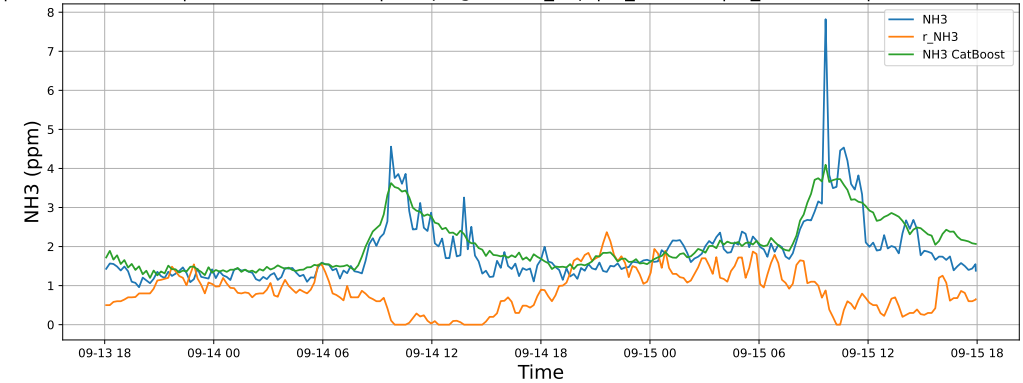
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

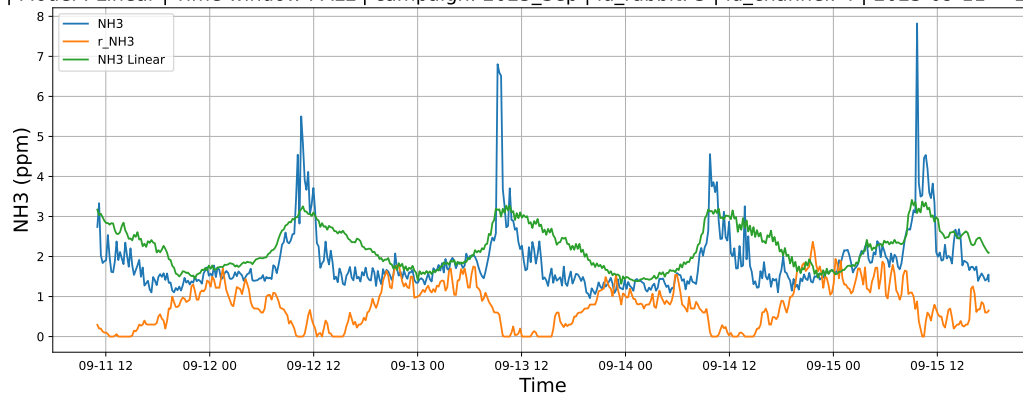


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

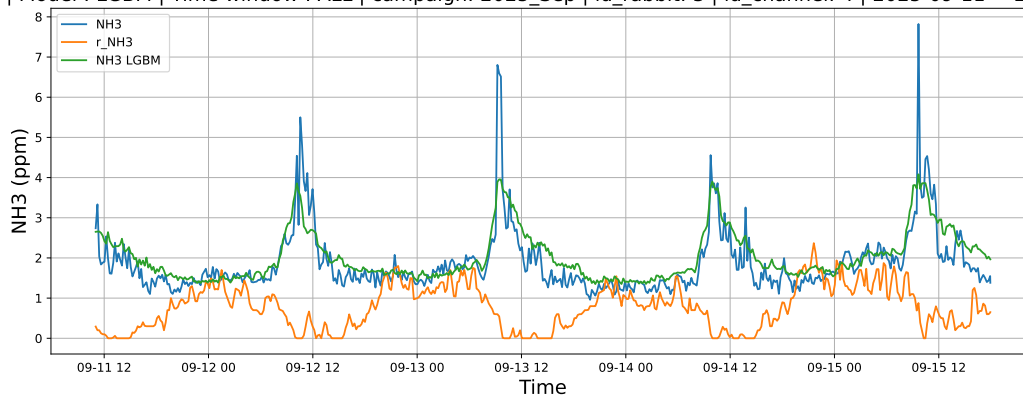


4.1.6.3 Time window : ALL

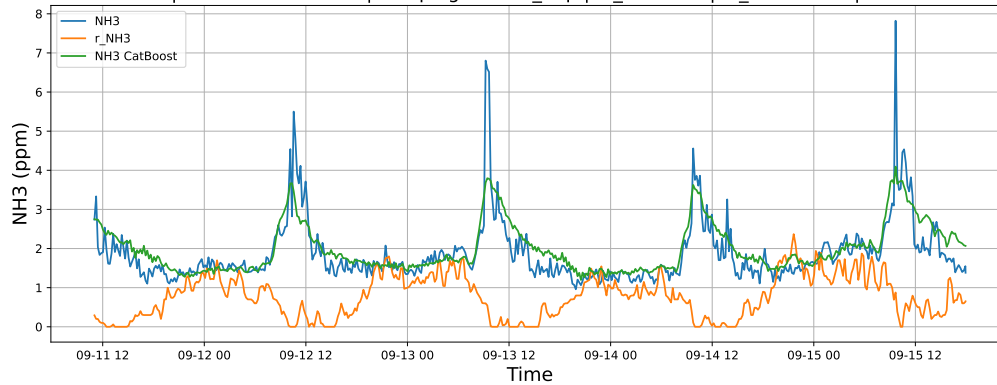
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



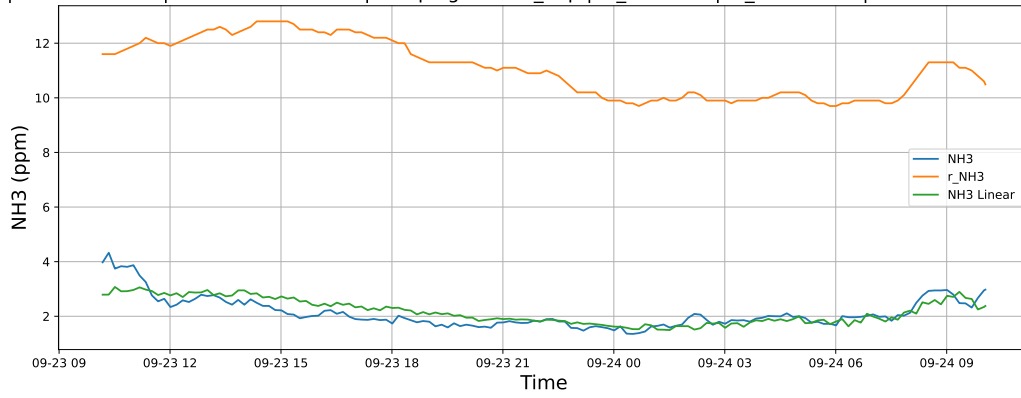
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



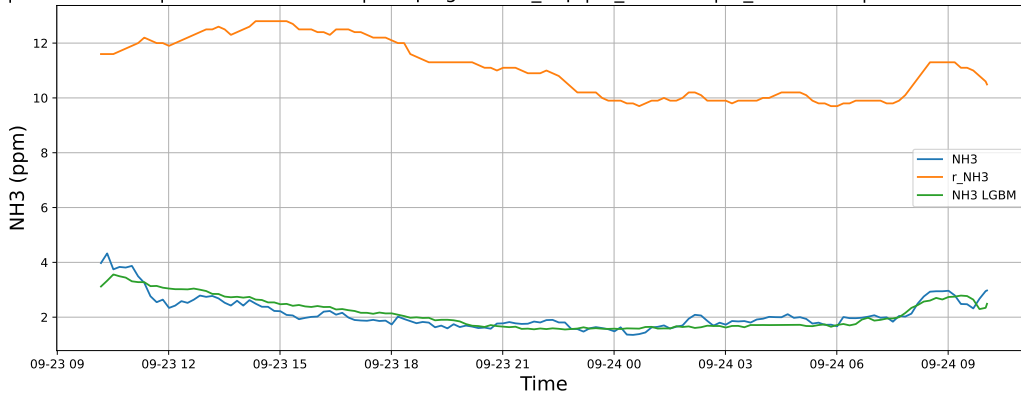
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

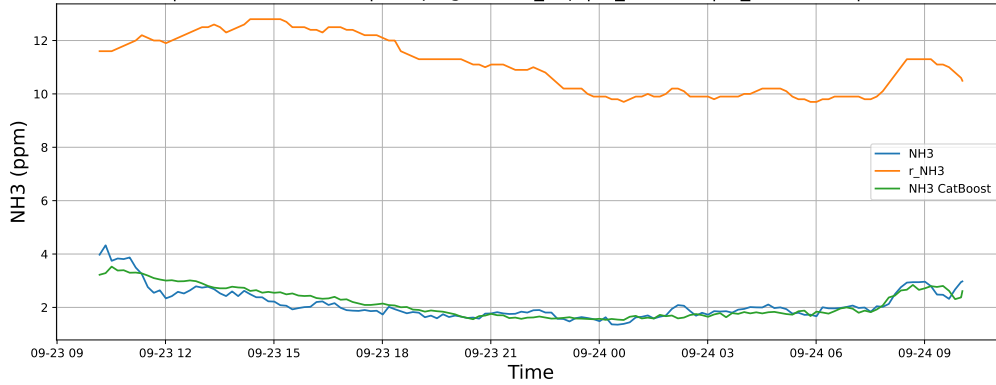
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

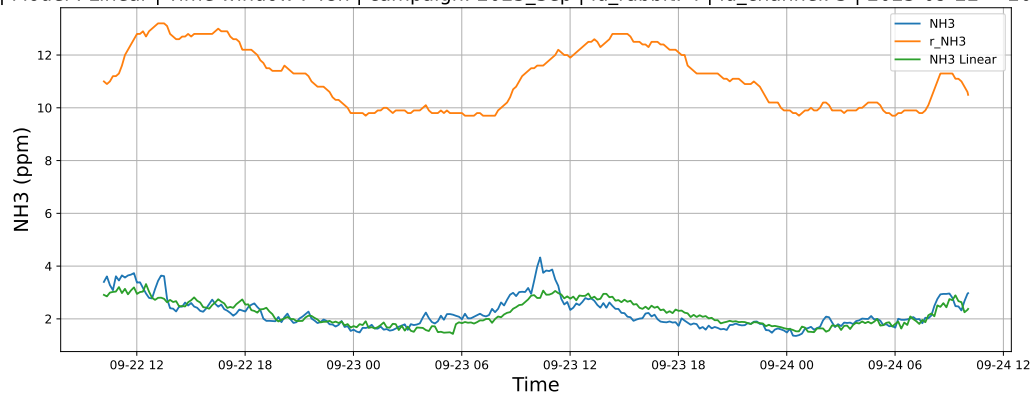


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

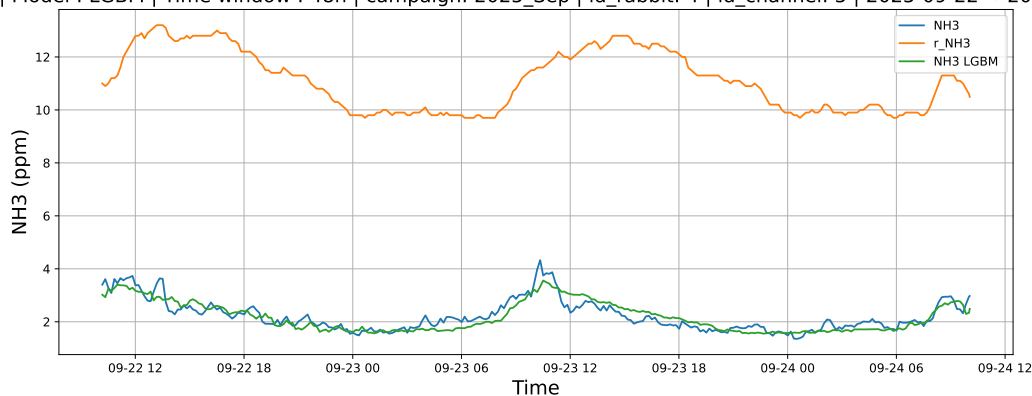


4.1.7.2 Time window : 48

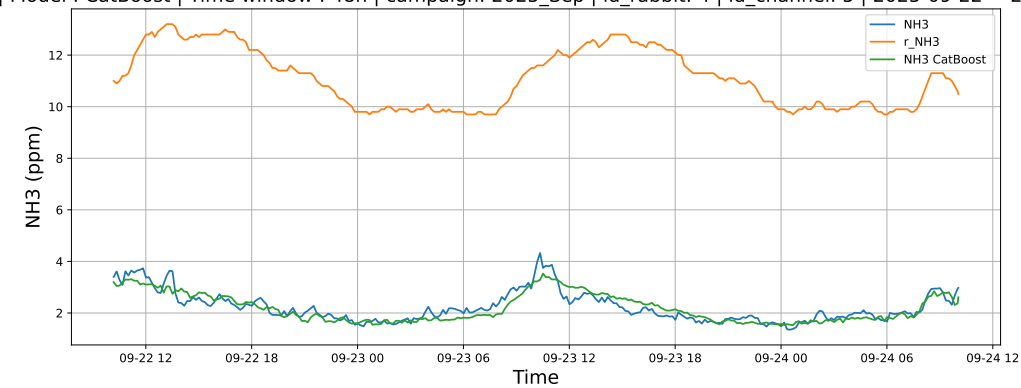
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

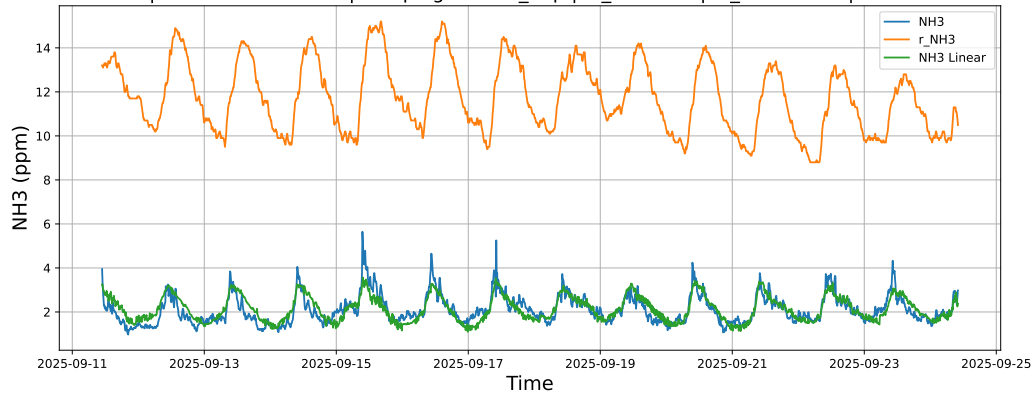


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

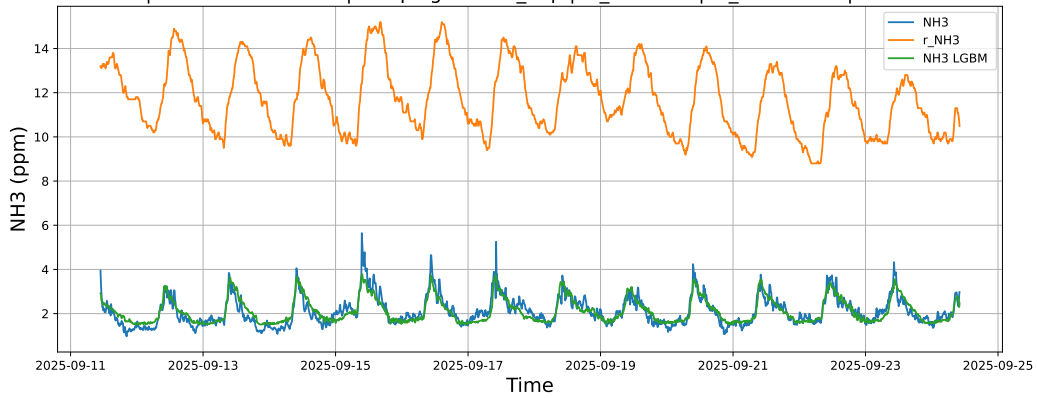


4.1.7.3 Time window : ALL

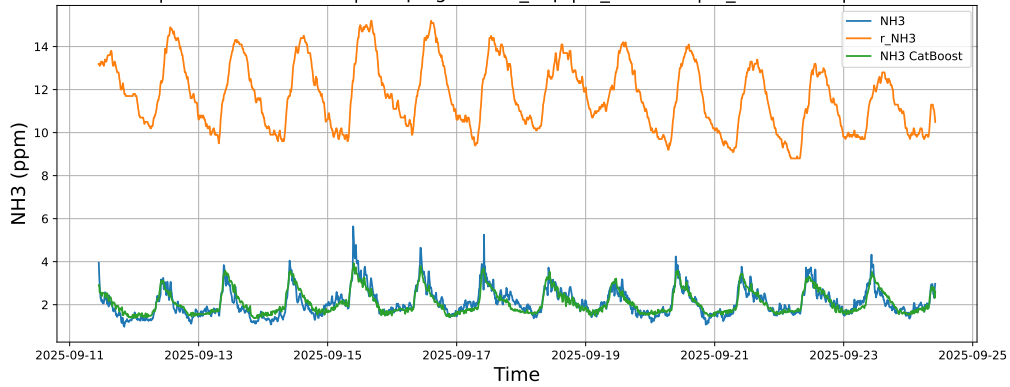
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



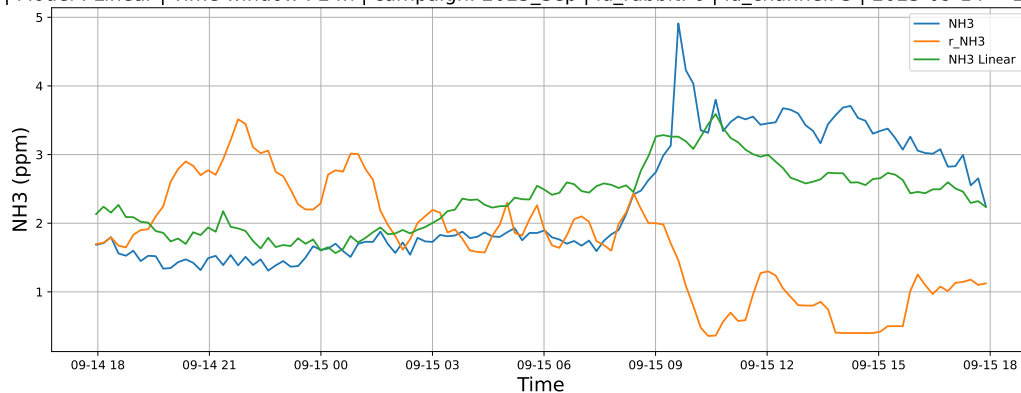
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



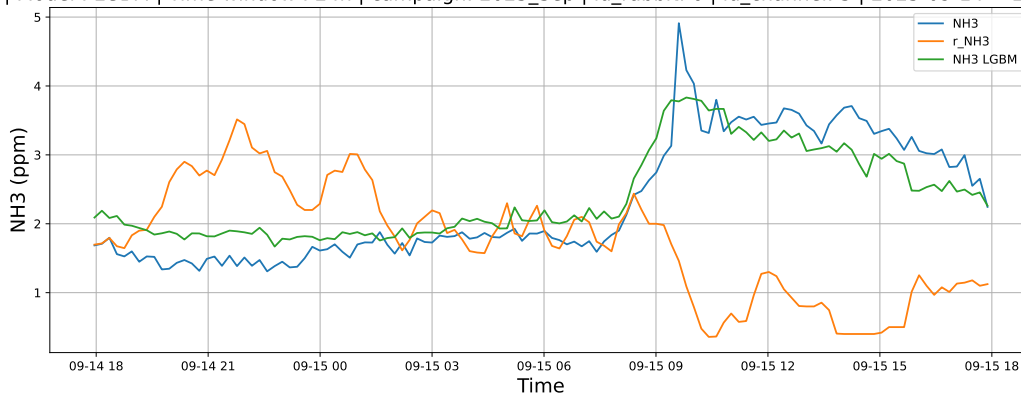
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

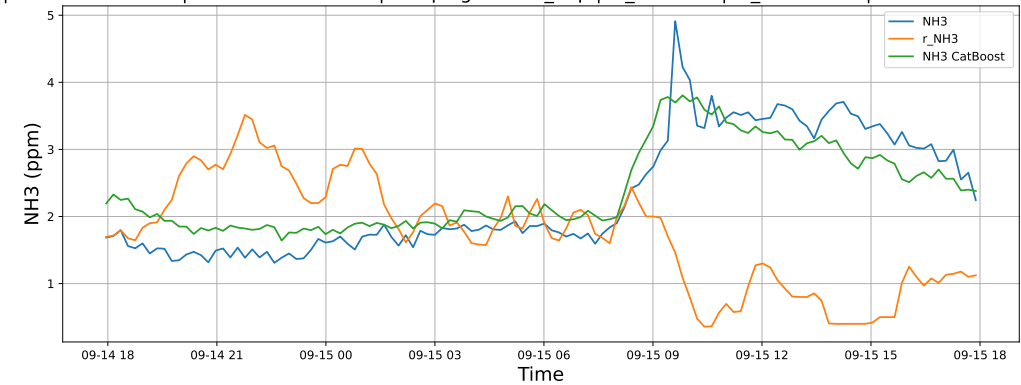
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

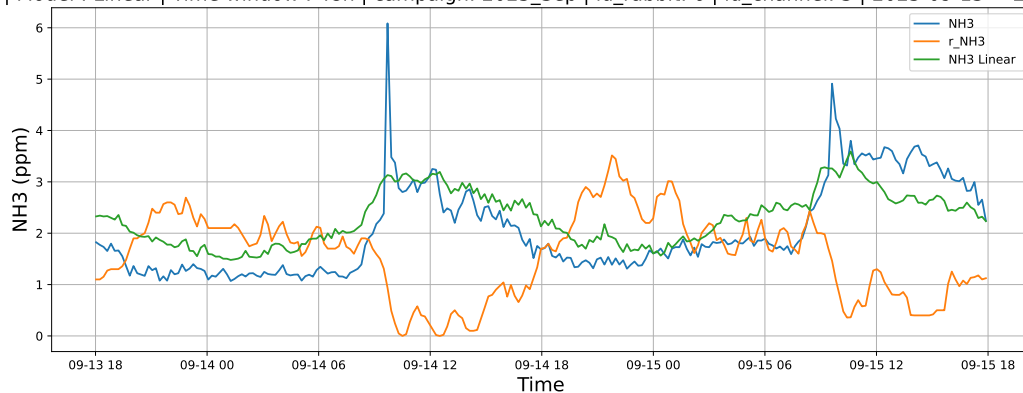


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

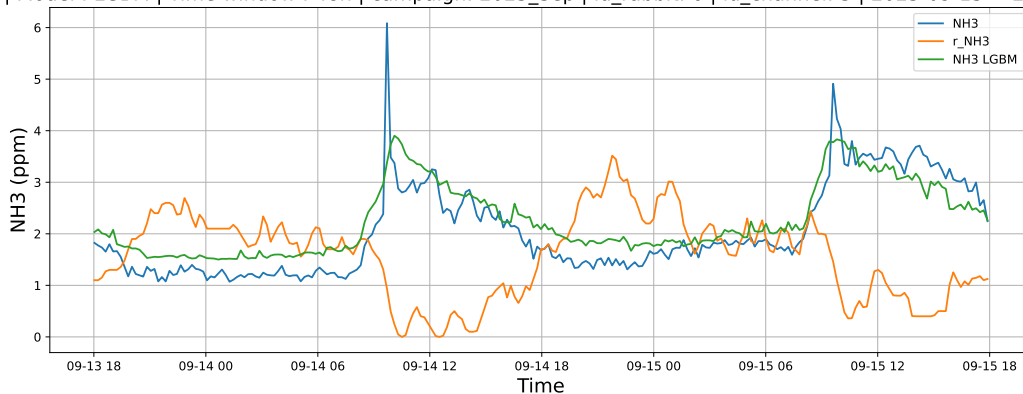


4.1.8.2 Time window : 48

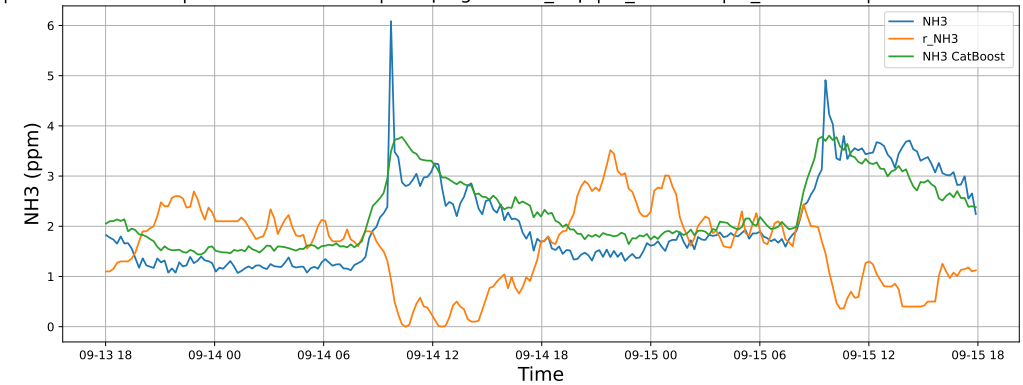
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

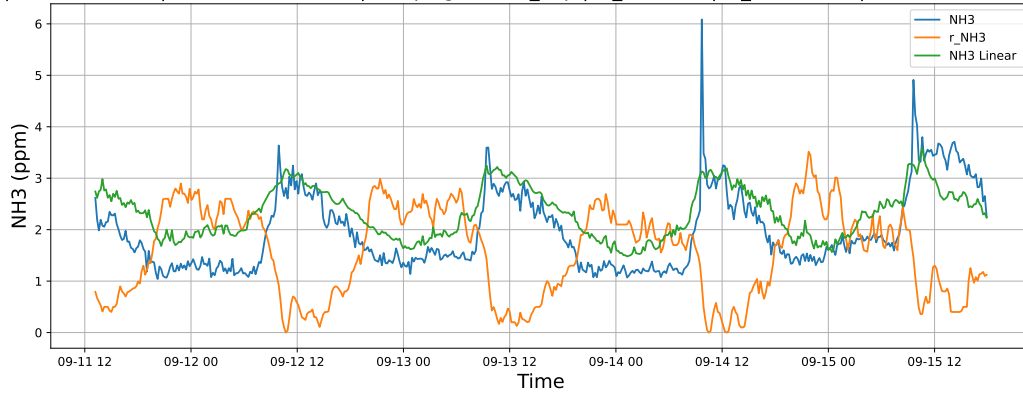


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

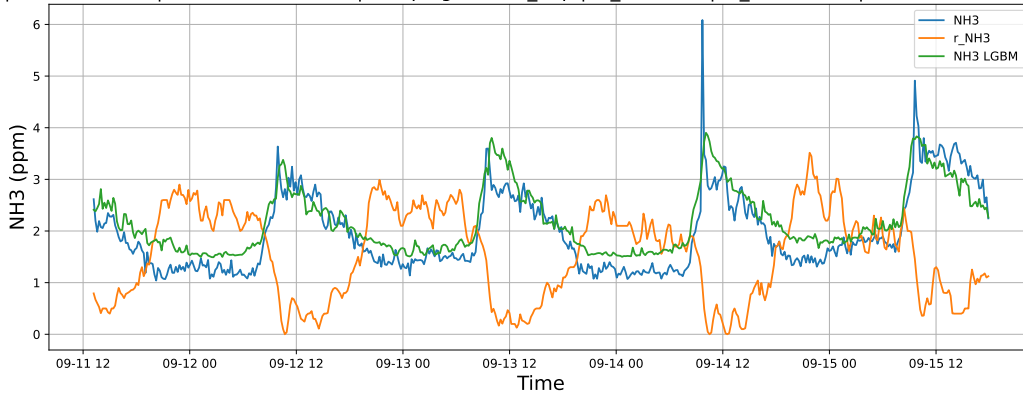


4.1.8.3 Time window : ALL

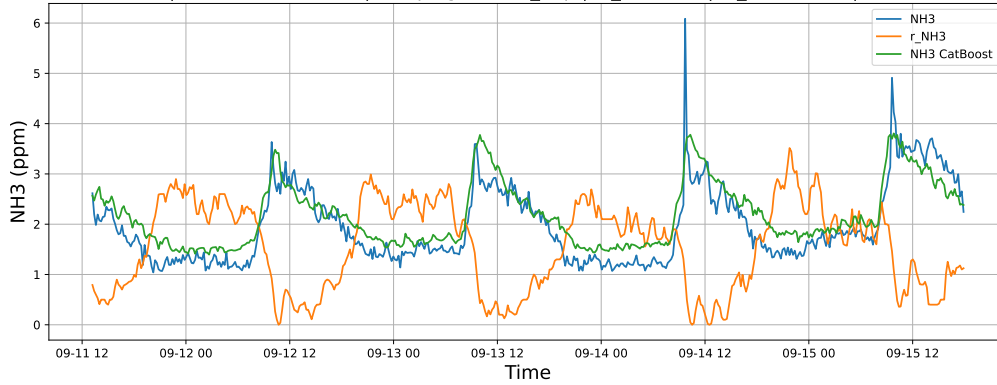
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

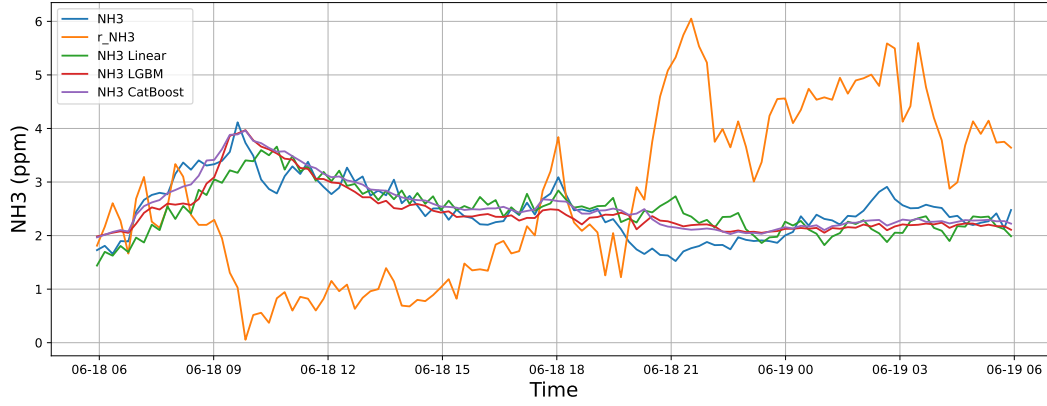


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

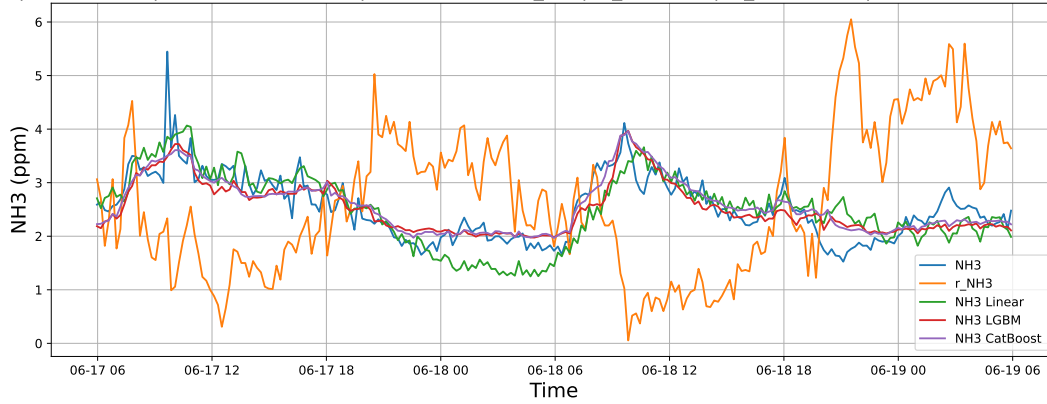
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



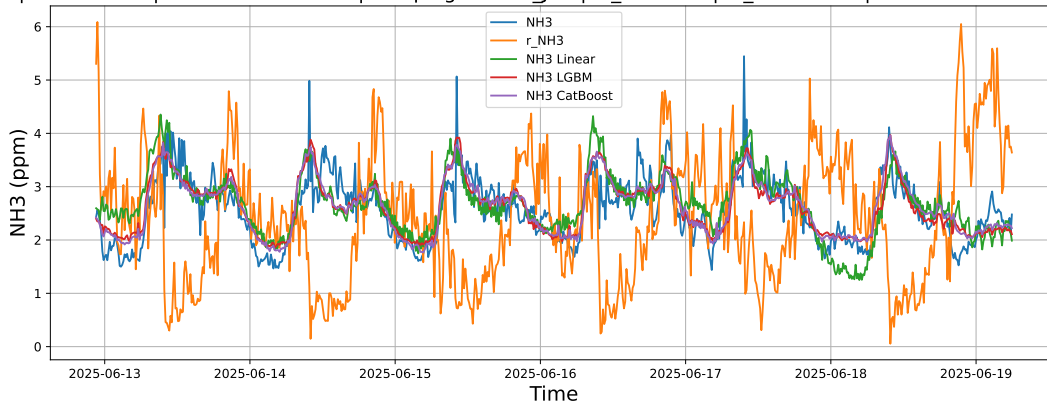
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

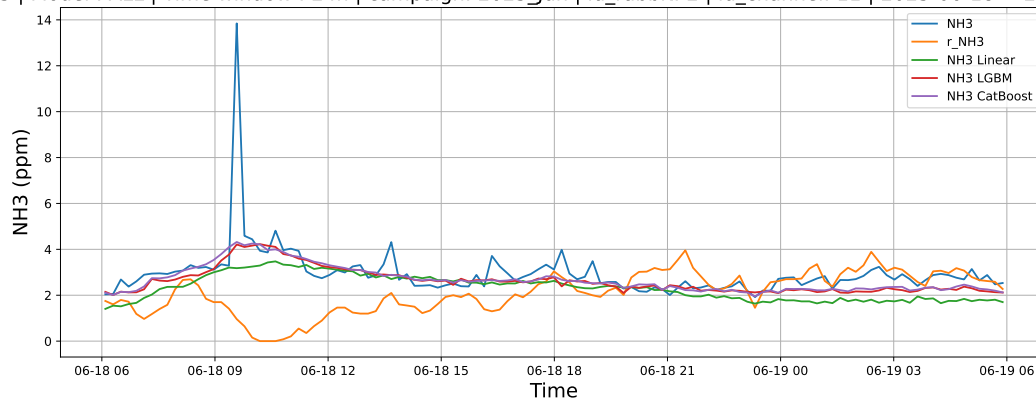
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

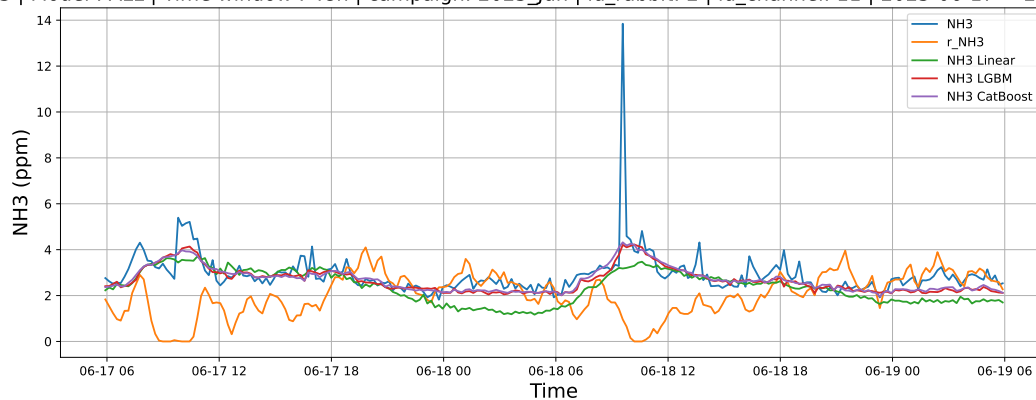
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



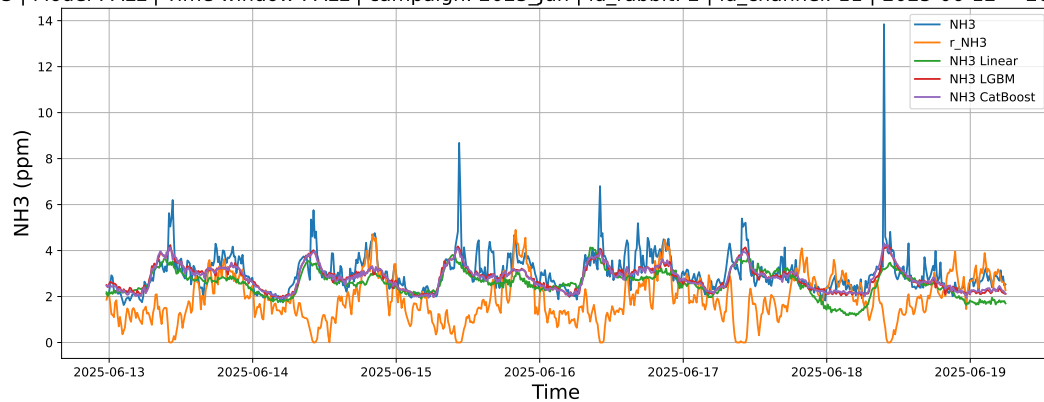
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

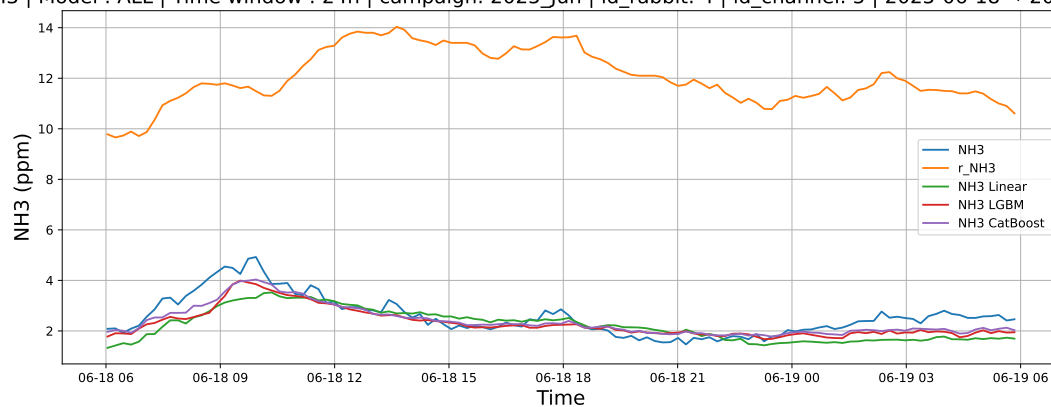
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

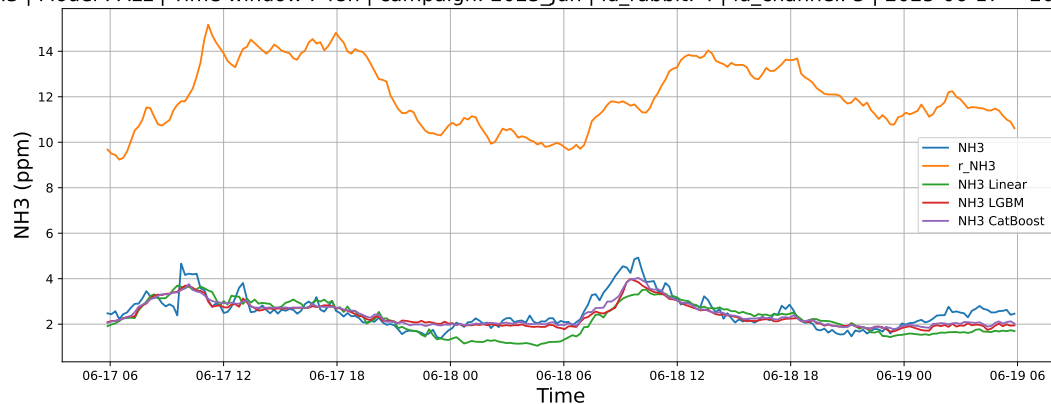
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



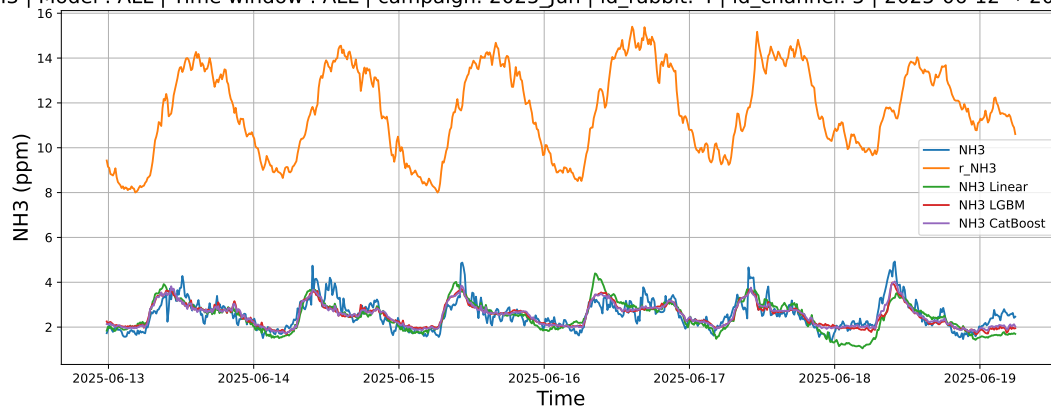
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

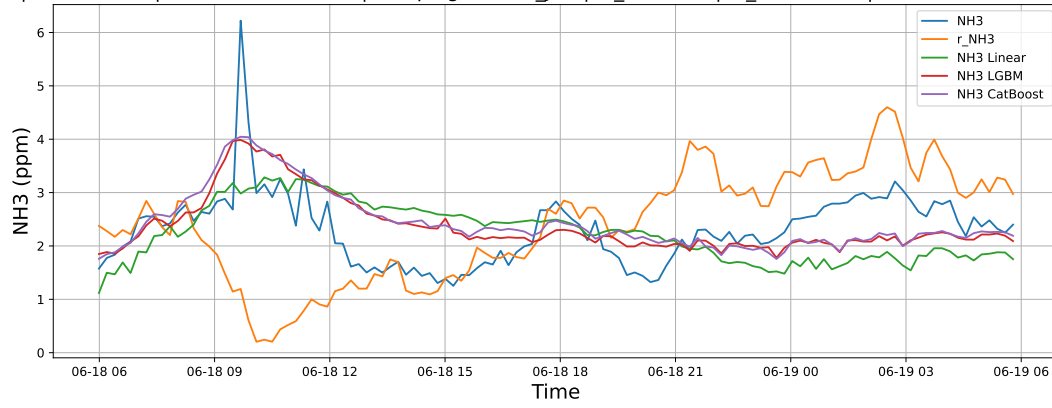
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

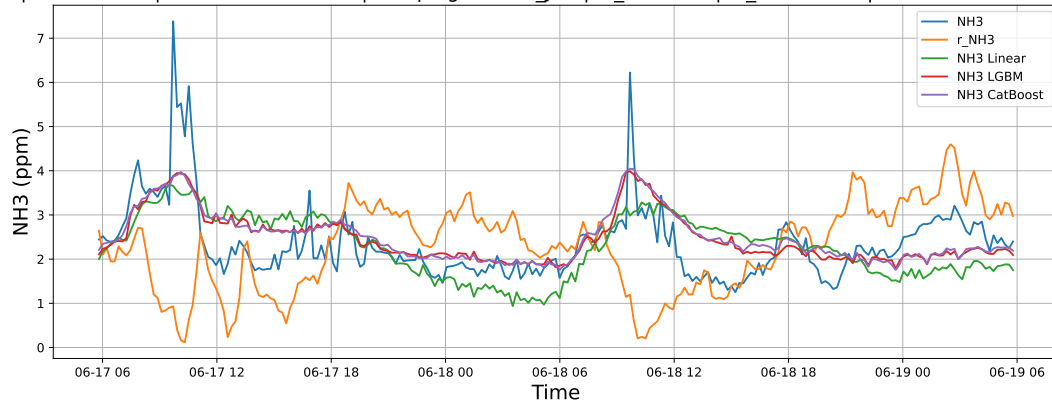
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



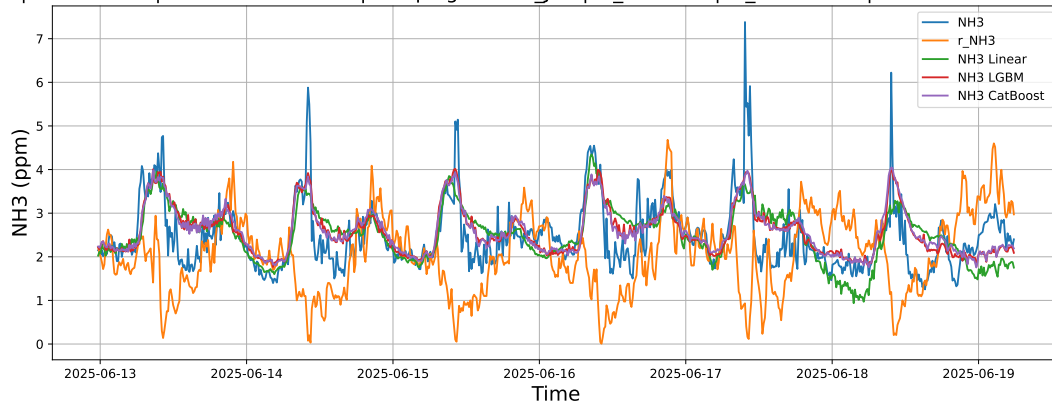
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

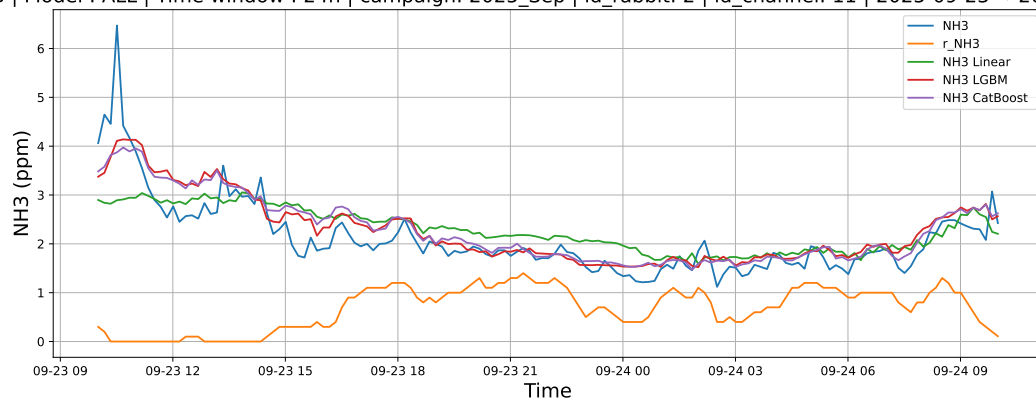
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

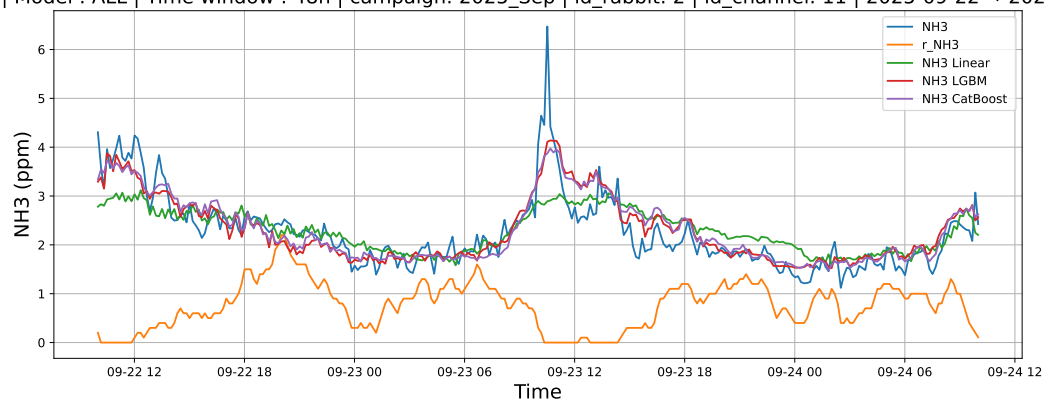
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



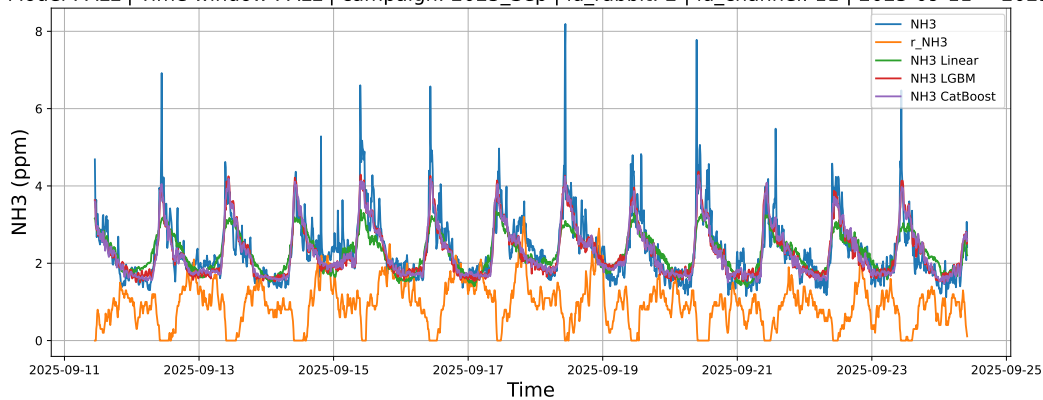
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

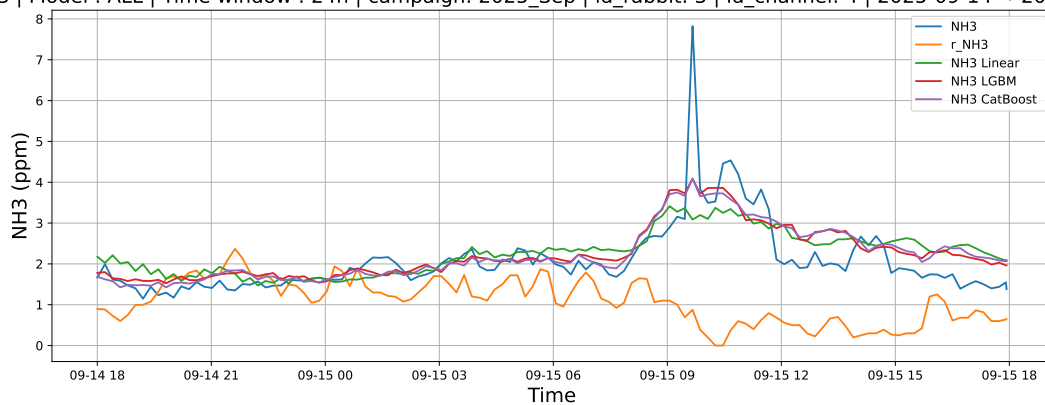
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

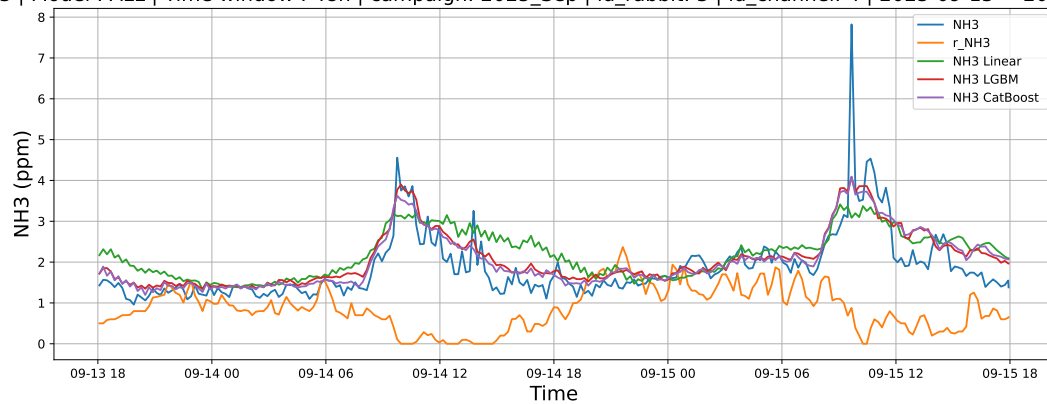
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



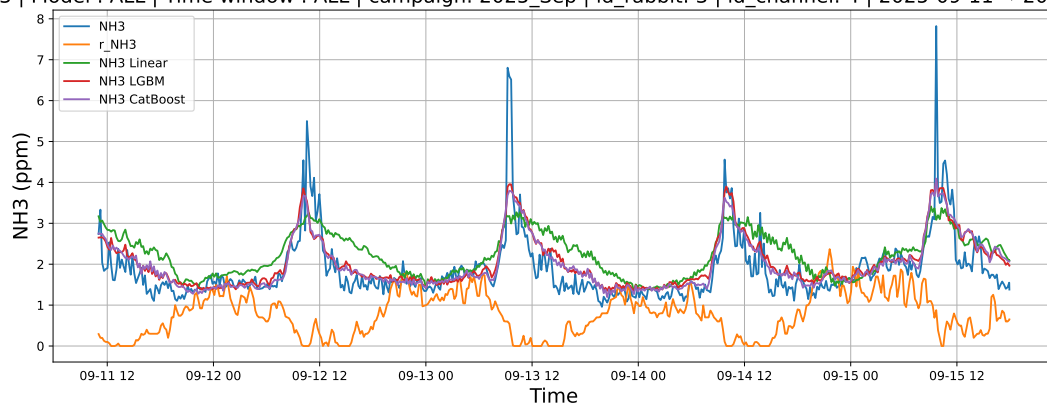
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

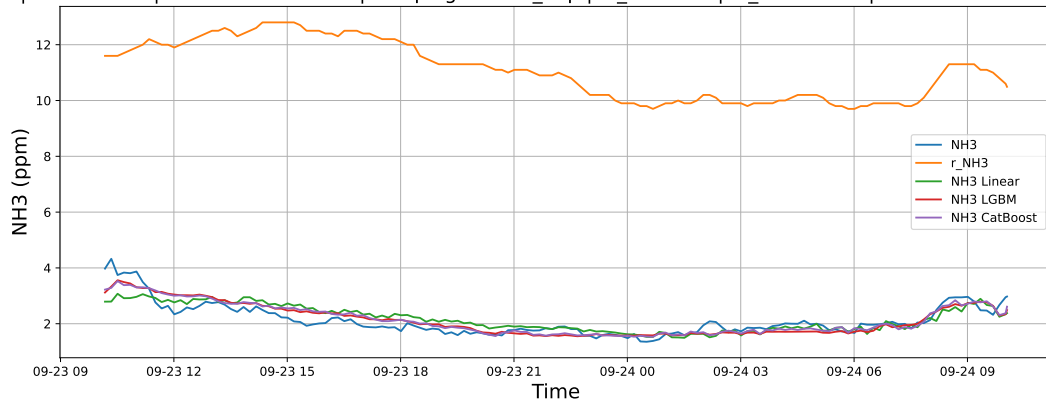
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

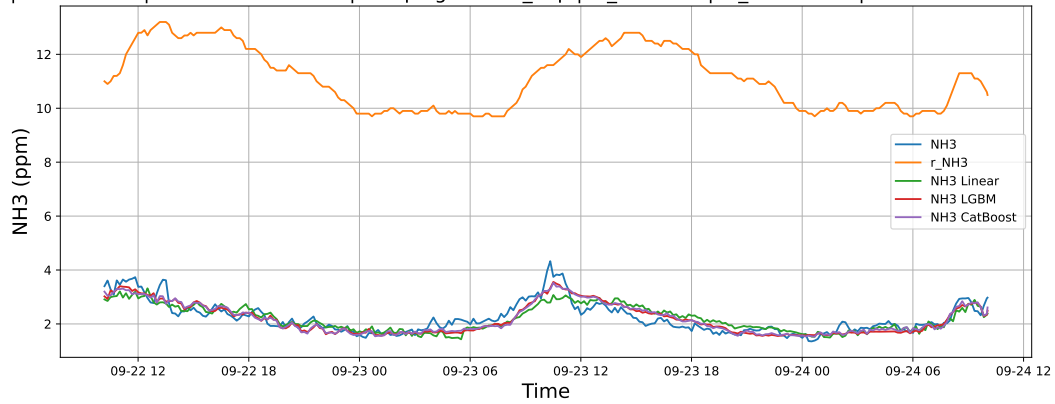
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



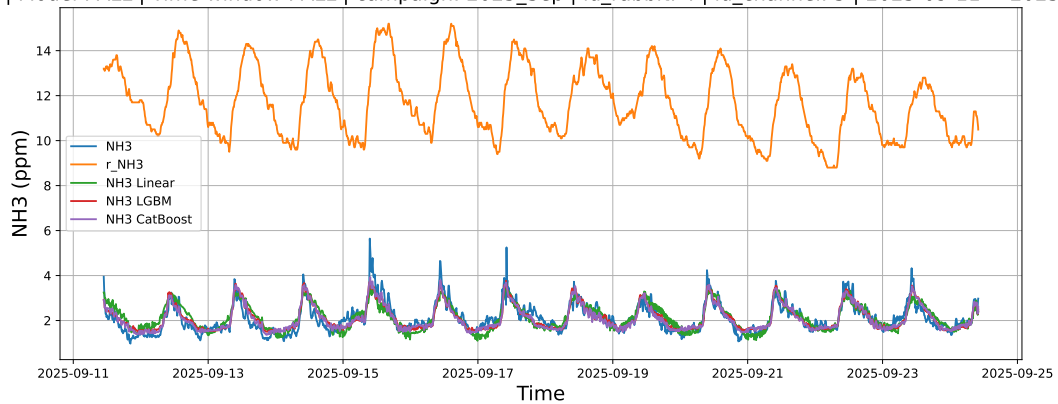
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

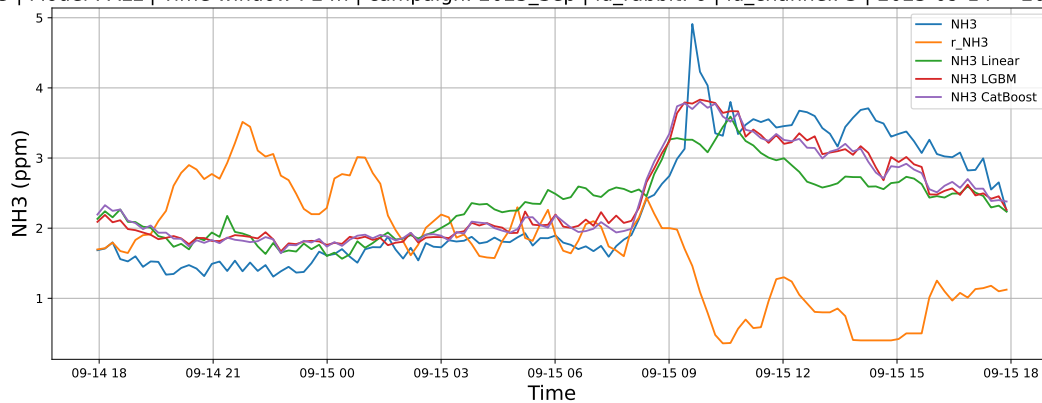
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

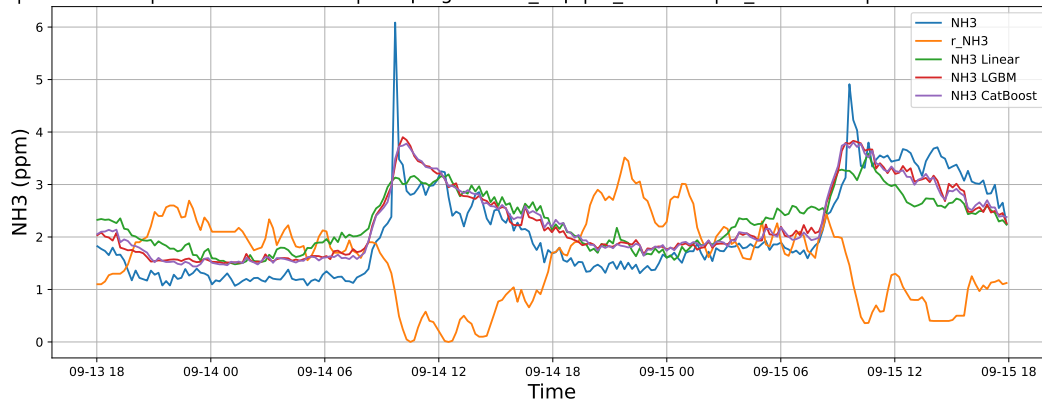
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

